scientific reports

Check for updates

OPEN

Unsteady CFD simulation of a rotor blade under various wind conditions

Sa. Kasmaiee[1], Si. Kasmaiee[1]✉ & A. Farshad[2]

Wind and gusts can significantly impact the performance of rotors and turbines. The transient behavior of the rotor should be carefully examined to account for these effects. This paper investigates the unsteady aerodynamic characteristics of a rotor blade under different wind conditions, such as direction, speed, and angle. A 3D transient Computational Fluid Dynamics (CFD) simulation using the dynamic mesh technique is performed to analyze the rotor dynamics. The unsteady Reynolds-averaged Navier–Stokes (URANS) equations with the *k-ω* SST turbulence model are solved. The rotor blade used for this study is the U15XXL Combo KV29 industrial blade, which has not been numerically analyzed before. The results show that wind in the same direction as the rotation reduces the thrust more than lateral or opposite wind. Lateral wind with a speed lower than 5 m/s decreases the blade performance, but higher speeds increase it. Higher lateral wind speeds also cause two peaks in the torque curve, forming a butterfly wing shape in the polar torque plot and multiple extrema in the torque curve with increasing speed. The maximum thrust shifts slightly to the left with increasing lateral wind speed. The lateral angle does not affect the average thrust produced in one blade revolution but only causes a spatial shift. The thrust production decreases as the angle approaches the opposite direction of rotation. The motion amplitude decreases, and the curve becomes smoother as this angle increases. A nearly straight line similar to no side wind is observed at 60 and 90 degrees, which is attributed to the constant effective angle of attack during the rotation cycle.

Keywords Rotor blade simulation, Incoming lateral and longitudinal wind, Unsteady aerodynamics, Transient simulation, Renewable energy, Computational fluid dynamics

Abbreviations

F_1	Blending function
k	Turbulence kinetic energy
P	Pressure
P_k	Production of turbulence
Re	Reynolds number
S	Area
t	Time
V	Velocity
V_∞	Free stream wind velocity
α	Angle of attack
α_0	Initial angle of attack
μ	Dynamic viscosity
μ_t	Eddy viscosity
ρ	Density
δ_{ij}	The kronecker tensor
τ	Surface stress
φ	Transport coefficients
ϕ, ψ	Angles between the direct wind and the axis of rotation blade
ω	Turbulence frequency length scale

[1]Department of Aerospace Engineering, Amirkabir University of Technology, Tehran, Iran. [2]Mechatronics Group, Department of Mechanical Engineering, Amirkabir University of Technology, Tehran, Iran. ✉email: siroos.kasmaiee@aut.ac.ir

Rotor and propeller performance in edgewise flight has attracted considerable attention recently due to the widespread use of fixed or variable-pitch propellers in modern air taxi and unmanned aerial vehicles[1–3]. Compared to fixed-wing aircraft, which are more suitable for long flights in open environments[4–6], multicopters have advantages for short duration missions in open or confined spaces due to their simplicity, maneuverability and vertical take-off capability[7]. The steady and unsteady flow phenomena around the rotor blade can be investigated through experimental[8] or various numerical methods. The numerical methods can be further classified into 1-D Blade Element Momentum (BEM) methods and 3-D Computational Fluid Dynamics (CFD) techniques. Recently, CFD analysis has been shown to predict the efficiency of hovering rotors[9] accurately. The hover condition is a critical design point because of its high power consumption and the challenge of predicting the strength and position of the vortex core[10]. The CFD technique can accurately handle a rotating blade's complex unsteady flow phenomena. Although it is computationally expensive, the CFD technique has been widely used as an effective analysis tool in academia due to the rapid advancement of High-Performance Computing (HPC) clusters[11]. According to the literature related to CFD, Lanzafame et al.[12] performed CFD analysis for performance prediction of the NREL Phase IV rotor. Bai et al.[13] developed a BEM model for small rotor blades, successfully validated through CFD analysis. Tong et al.[14] studied the effects of blade airfoil chord length and rotor diameter on the aerodynamic performance of straight-bladed vertical-axis turbines. Yoon and Lee[15] investigated the aerodynamic parameter analysis and design of a quadrotor air vehicle in hover. Their results indicated that optimized twist angle and twist-starting position enable maximum hovering thrust in the proposed quad-copter. Lei and Wang[16] studied the aerodynamic characteristics of a micro quadrotor aircraft considering the horizontal wind disturbance. They showed that the rotors' thrust increases gradually with wind speed, resulting from increased pressure difference between the upper and lower surfaces. Cai et al.[17] simulated the unsteady aerodynamic configuration of a full-scale horizontal axis wind turbine with consideration of wind shear, tower shadow and yaw motion. Their results indicated that each blade simultaneously reaches its maximum and minimum aerodynamic loads during the rotation cycle. Al-Ghriybah et al.[18] studied the effect of the inner blade position on the rotor's performance to determine the inner blade's best position. The behavior of the blade tip vortices of a turbine equipped with a brimmed diffuser shroud was investigated by Takahashi et al.[19]. Their simulation results indicated that a pair of vortices consisting of a blade tip vortex and a counter-rotating vortex was generated between the blade tip and the inner surface of the diffuser. The rotor was evaluated for steady and unsteady aerodynamics analysis under surge motion[20], yawed gusts[21], ground effect[22,23], and variable pitch[24]. Zhang et al.[20] demonstrated that the mean power of the twin-rotor is increased by up to 13% under surge motion compared to the single rotor and that the small spacing ratio enhances the power gain. They revealed that surge motion significantly impacts the wake interaction between the twin rotors and facilitates wake recovery. Cai and Gunasekaran[21] examined the propeller performance under steady and sinusoidally time-varying freestream (stream-wise or longitudinal gust). They reported that the propeller thrust, power, pitching, and rolling moment increase with the incidence angle at the same advance ratio. Tanabe et al.[23] simulated a variable-pitch-controlled quadrotor drone using a high-fidelity CFD solver in the ground effect. They detected a flow recirculation for the quadrotor drone at an intermediate height of approximately 0.5 times the rotor diameter. Also, they observed that the required power of the quadrotor drone varies significantly with the rotor height from the ground and that a local power increase occurs around a rotor height of 0.5D. Nowadays, the use of artificial intelligence is being considered in all branches of engineering[25–27]. In the literature, studies have been conducted to improve the efficiency of rotors by optimizing blade shape[28,29] or using flow control methods[30–32]. The effect of the functional parameters of the blowing and suction jets on pitching airfoils in dynamic stall conditions was discussed by the authors in[32–34]. The authors[30] recently optimized the blowing jet performance for a dynamic stall airfoil. They found that the optimal jet location was near the leading edge vortex formation, about 2 to 5 percent of the chord. The jet angle of 55 to 70 degrees improved the aerodynamic performance better. Salah et al.[35] studied the performance of dual-rotor wind turbines (DRWT). They show that the DRWT significantly affects the efficiency of wind energy.

Considerable research has been carried out on fixed and variable-pitch rotor blades, focusing on commercial blades featuring symmetrical blades like Naca0012 or Naca0015. Several studies have also been conducted on the NREL rotor. However, the authors did not find a study that works on the Combo KV29 rotor. On the other hand, most of the works investigated the steady state of the flow, and limited studies investigated the unsteady flow behavior. In the unsteady flow studies, limited cases were investigated due to the high computational time; for example, Ali and Kim[11] only investigated 3 cases of wind condition (without tilt, with tilt -20, and with yaw -20). On the other hand, numerical modeling of rotor aerodynamics and performance under wind poses several challenges. Therefore, the resulting rotor requires more in-depth studies for performance evaluation under different operational conditions. This study examined this rotor numerically under different winds for the first time. In addition, we tried to consider all the parameters and conduct a comprehensive study on wind's effect on the rotor's performance. In total, 16 different scenarios were considered for wind speed and direction, and a comprehensive study was conducted on the effect of wind on the performance of this rotor. Three-dimensional computational fluid dynamics was employed for this investigation. The motion is modeled as a dynamic mesh and the flow is solved unsteadily to investigate the physics of the flow more accurately. In the following sections, the governing equations of the problem are presented first. Then, the modeling approach is described, and after validating the solution and ensuring its accuracy, the performance of the rotor is examined in the presence of wind. Several wind scenarios are considered, and the effect of wind direction, angle, and speed on thrust and rotor torque is investigated. Flow physics is studied by examining contours and isosurfaces.

Numerical method

Governing equations

Unsteady Reynolds averaged Navier–Stokes (URANS) equations are solved through commercial CFD software Ansys Fluent. Air was considered an incompressible and Newtonian fluid. Thus, the mass conservation equation is as follows[36,37]:

$$\frac{\partial \overline{\boldsymbol{u}}}{\partial \boldsymbol{x}} + \frac{\partial \overline{\boldsymbol{v}}}{\partial \boldsymbol{y}} + \frac{\partial \overline{\boldsymbol{w}}}{\partial \boldsymbol{y}} = 0 \tag{1}$$

Considering that $\overline{V} = (\overline{u}, \overline{v}, \overline{w})$, so, linear momentum conservation equations can be written as below:

$$\frac{\partial \overline{V}_i}{\partial t} + \overline{V}_j \frac{\partial \overline{V}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \overline{P}}{\partial x_i} + \frac{\partial}{\rho \partial x_j}\left(2\mu \overline{e}_{ij} - \rho \overline{V_{i'} V_{j'}}\right) \tag{2}$$

In this equation, $\overline{e}_{ij}$ is the strain tensor and is determined from the following relation:

$$\overline{e}_{ij} = \frac{1}{2}\left(\frac{\partial \overline{V}_j}{\partial x}_i + \frac{\partial \overline{V}_i}{\partial x}_j\right) \tag{3}$$

According to the Boussinesq assumption, the Reynolds stress can be obtained from the following equation:

$$\tau_{ij} = -\rho \overline{V_{i'} V_{j'}} = 2\mu_t \overline{e}_{ij} - \frac{2}{3}\rho k \delta_{ij} \tag{4}$$

where μ_t is the eddy viscosity, k is the mechanical energy of turbulence, and δ_{ij} is the kronecker delta. A hybrid k–ε and k–ω Shear Stress Transport (SST) turbulence model expanded by Menter[38] is applied for accurate CFD modeling of the rotor. The SST model is considered one of the best turbulence models for rotor applications based on computational accuracy[11]. In this model, two equations of turbulent kinetic energy and specific dissipation rate are added to the system of equations for eddy viscosity modeling as follows:

$$\frac{\partial \mathrm{k}}{\partial t} + \overline{V}_j \frac{\partial \mathrm{k}}{\partial x_j} = P_k - \beta^* \omega k + \frac{\partial}{\rho \partial x}_j \left[(\mu + \sigma_k \mu_t)\frac{\partial \mathrm{k}}{\partial x_j}\right] \tag{5}$$

$$\frac{\partial \omega}{\partial t} + \overline{V}_j \frac{\partial \omega}{\partial x}_j = \alpha S^2 + \frac{\partial}{\rho \partial x}_j \left[(\mu + \sigma_\omega \mu_t)\frac{\partial \omega}{\partial x}_j\right] + 2(1 - F_1)\frac{\sigma_{\omega_2}}{\omega}\frac{\partial \mathrm{k}}{\partial x_j}\frac{\partial \omega}{\partial x}_j - \beta \omega^2 \tag{6}$$

F_1, P_k, α, β, β^*, S, σ_k,σ_{ω_2} and σ_ω are closure parameters.

Simulation set-up

U15XXL Combo KV29 industrial rotor blade was used. This is an industrial blade produced by T Motor Company. This engine is used in unmanned vehicles such as quadrotors. The reason for using it in this study was that the main and final goal is designing a quadrotor that is capable of transporting hospital equipment and supplies so that transportation in the hospital can be done more easily and quickly. For this reason, this industrial rotor was used, and aerodynamic analyses were performed on it in different conditions to design its control system. This rotor has features such as considerable power, a matched propulsion solution, stability, and reliability that are necessary for our quadrotor. The experimental model of this rotor was prepared, and then a computer-aided design (CAD) model of the rotor was created using Solidwork software. The CAD model was generated from a mesh file obtained by 3D laser scanning of the rotor. A view of this model can be seen in Fig. 1. This blade's length, width, and depth are 62, 6.51, and 2.16 inches, respectively.

In this study, a second-order two-dimensional finite volume method was applied, and a SIMPLE algorithm was utilized to couple pressure–velocity. Air density was considered as an ideal gas. The turbulent property was modeled using URNAS models. The k-ω SST was used because of its reported accuracy in previous studies[11]. The discretization method adopted the second order upwind scheme. The internal iteration of 20 iterations each time was selected. The simulation convergence was checked by monitoring residuals less than 10^{-6}. We considered

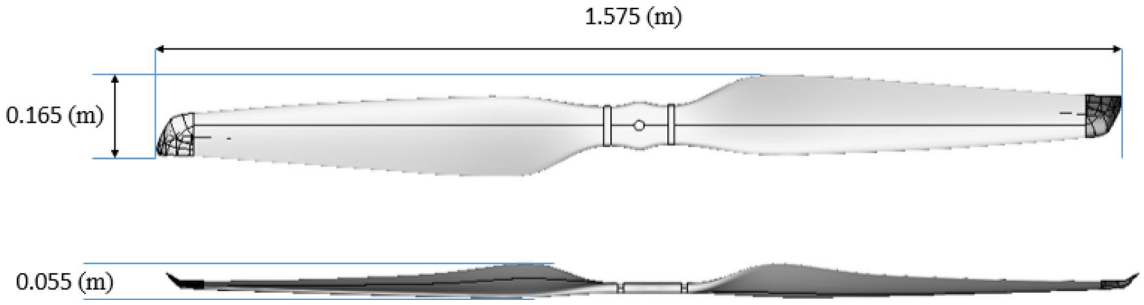


Figure 1. CAD model of U15XXL Combo KV29 blade.

the time step to be small, and this caused each step to have little change compared to the previous step, and 20 repetitions are enough to converge to 10^{-6}. Because we set the condition as 10^{-6}, many times, it did not even reach 20 repetitions, and it was left out of the loop after 10 to 15 repetitions.

Computational domain & boundary conditions

Considering that the dynamic mesh model is supposed to be used, we must define the rotating computational zone near the model and the stationary zone in the rest. Therefore, the calculation area should be divided into two parts according to Fig. 2. In order to be able to use this computational area in the dynamic mesh method, the rotating zone is designed as a cylinder. The rotating zone is twice the length of the blade. The length of the stationary zone from the back of the blade is about 20 times the length of the blade, and from the front, it is 6 times the length of the blade. In Section "Domain independency", the domain independence is examined to ensure the sufficiency of the domain size. The boundary condition of the stationary zone's side walls was considered a pressure outlet, the cross section near the blade was assumed as inlet velocity, and the end section of the cylinder was considered a pressure outlet. The wall with the no-slip condition was also considered on the blade. The boundary surface of the inner cylinder was also considered as the interface of two zones. In Fig. 2, computational domain and boundary conditions are specified.

The mesh was created using the commercial software Pointwise. Structured and unstructured hybrid grids were used for meshing. On the surface of the blade, the main part of the blade was a meshed structure, but the tip of the blade and its hub were meshed in a triangular shape. The boundary layer was meshed as 10 cubic structured layers, and according to the selected turbulence model, the size of the first layer was considered such that y^+ of this layer is about one. Except for the first ten layers related to the boundary layer, the mesh inside the computational zone was also considered unstructured. The T-rex feature of pointwise software has been used to build this mesh. In Fig. 3, a sample of the mesh inside the moving and fixed cylinder and the structured mesh

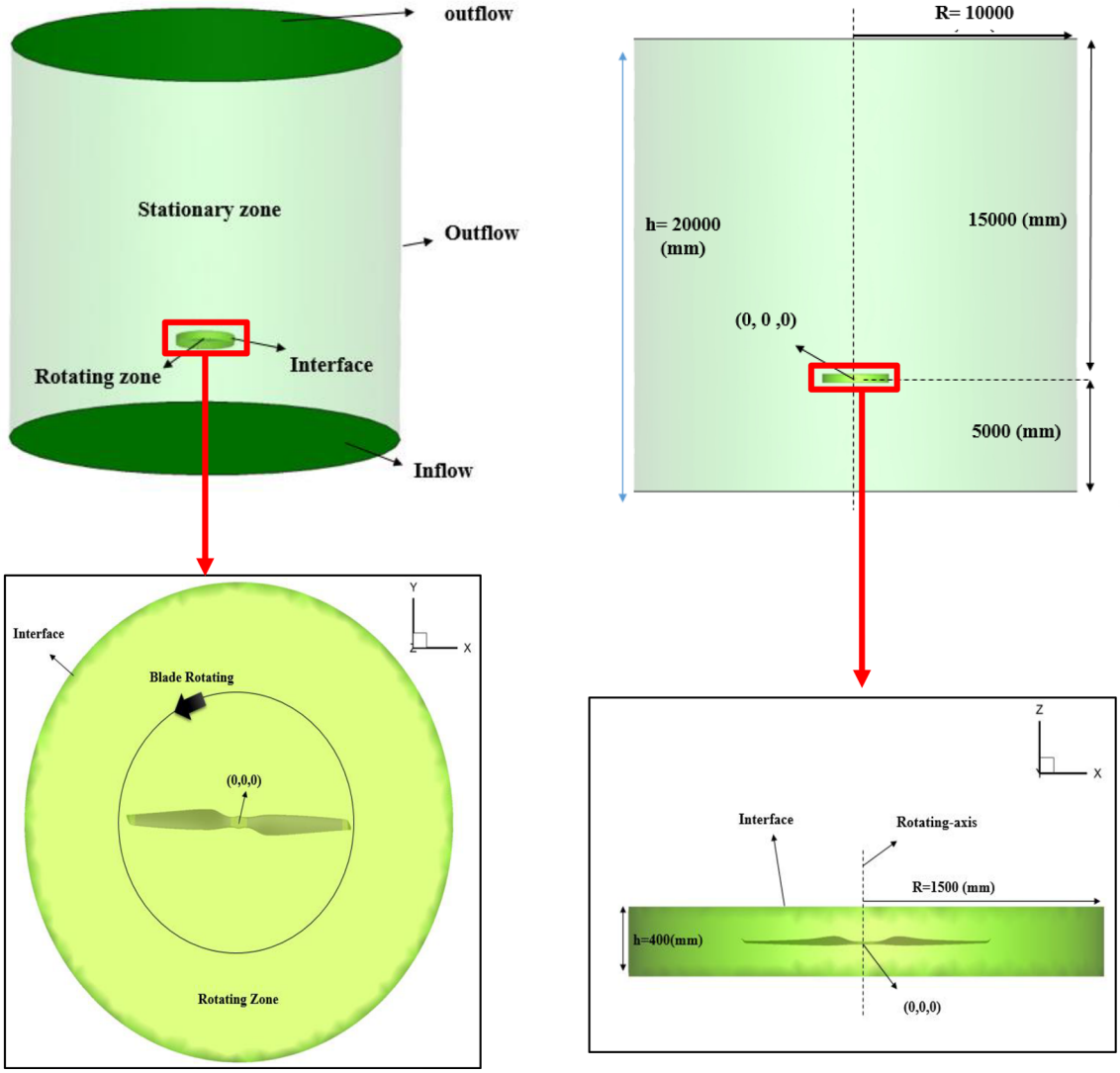


Figure 2. Computational domain and boundary conditions of the zones.

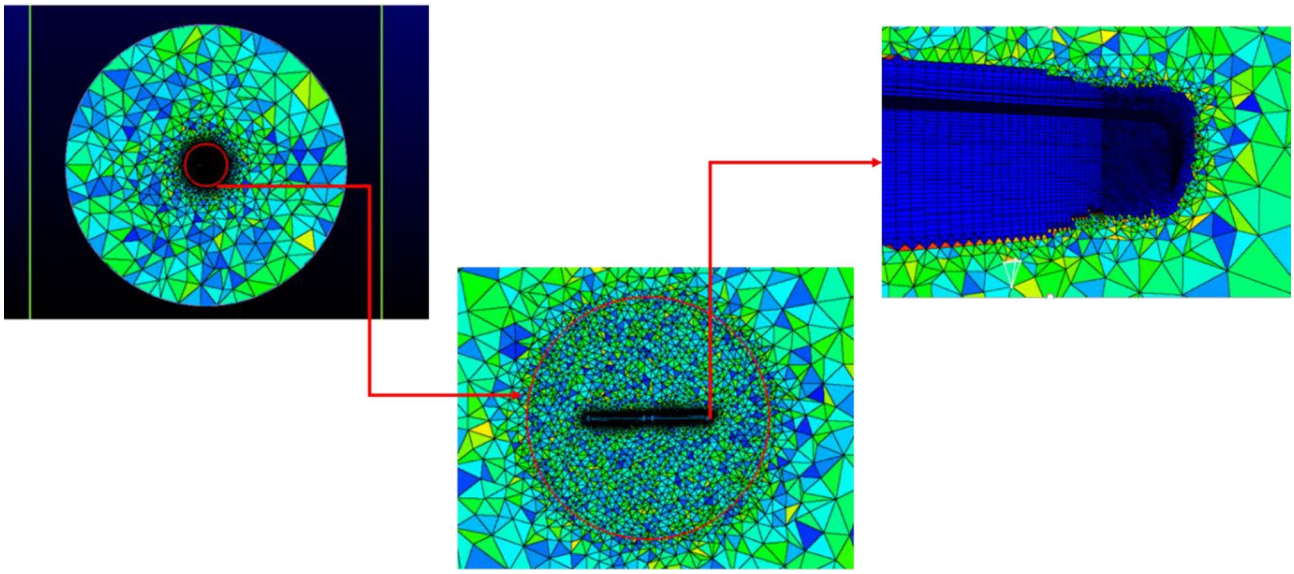


Figure 3. Computational grid in the rotating and fixed zones and a view of the generated boundary layer mesh in $z = 0$.

of the boundary layer can be seen. This is a cross-sectional view of the grid in the z = 0 plane. In Fig. 4, the final mesh used on the blade's surface can be observed. As can be seen, the hub and tip are meshed triangular and unstructured, and the main part of the blade is meshed structured.

Numerical verification

Mesh independency

In order to check the independence of the results from the number of grids and the consistency of the parameters with increasing the number of cells, 4 grids were examined. Considering that we have three effective areas (surface of blade, rotating area and fixed area), which can be seen in Fig. 5, the gridding has been done based on changing the number of elements in these three areas and the number of cells on these surfaces as well as the layer between these surfaces has changed. The number of cells and the total size of the investigated meshes can be seen in Table 1.

In order to check the independence of the results from the grid size, thrust force, and torque were examined for different grids at a fixed rotor speed. The rotor rotational speed was considered equal to 896 rpm. In the continuation of the present study, we checked the effect of the wind at this speed. We chose it because we wanted to carry out most of the flights with 30% engine power of this blade. This study's main and final goal is to design a quadrotor capable of transporting hospital equipment and supplies so that hospital transportation can be done more easily and quickly. The operational throttle for this mission was set at 30% throttle. The time step was considered equal to 0.0002 s, and the results were averaged for a complete blade revolution (7th revolution). The results of thrust and torque were calculated for different grids, as seen in Table 2. The relative error of each grid to the previous grid is also provided for a more detailed analysis. As can be seen, the relative error for Mesh 3 is low with respect to the next grid, so this grid is suitable. On the other hand, since the final limit of angular speed was 2111 rpm, to ensure the adequacy of the mesh at this speed, the last two meshes were examined, presented in Table 3. We did not have the mesh size independent of the lower rotational speed for the first and second mesh, so it was pointless to simulate these meshes again at high rotational speed. However, to ensure that the selected mesh (Mesh 3) is also sufficient at high rotational speed, we simulated the last two meshes to show that the selected mesh (Mesh 3) is also sufficient in these conditions. Considering the low relative error of these two meshes, this number of elements is also sufficient at this speed.

Time independency

In order to solve the unsteady flow, it must first be determined how many cycles (rotation of one complete revolution of the blade) are needed for the simulation to converge. In fact, we continue the solution until the unsteady solution in a blade motion cycle has little difference from the previous cycle. For this purpose, mesh 3 (one million and seven hundred thousand cells) with a time step of 0.0001 s was considered. Considering that a maximum of 90% throttle is used in real flight, we consider the same engine speed for checking. This engine speed is equal to 2111 rpm. According to the diagram of thrust and torque versus the number of cycles for this engine revolution in Fig. 6, from the 5th cycle onwards, the results have not changed much with respect to the subsequent cycles. Therefore, cycles 6 to 7 were considered as the solution of the transient state in this simulation, and all simulations continued until cycle 7.

The independence of the time step of the solution should be checked. Considering that the case with 2111 rpm is selected for examination, by trial and error, 4 time steps of 0.0001, 0.0002, 0.0004 and 0.001 s were considered. In order to more accurately check the relative error of each time step compared to the previous time step

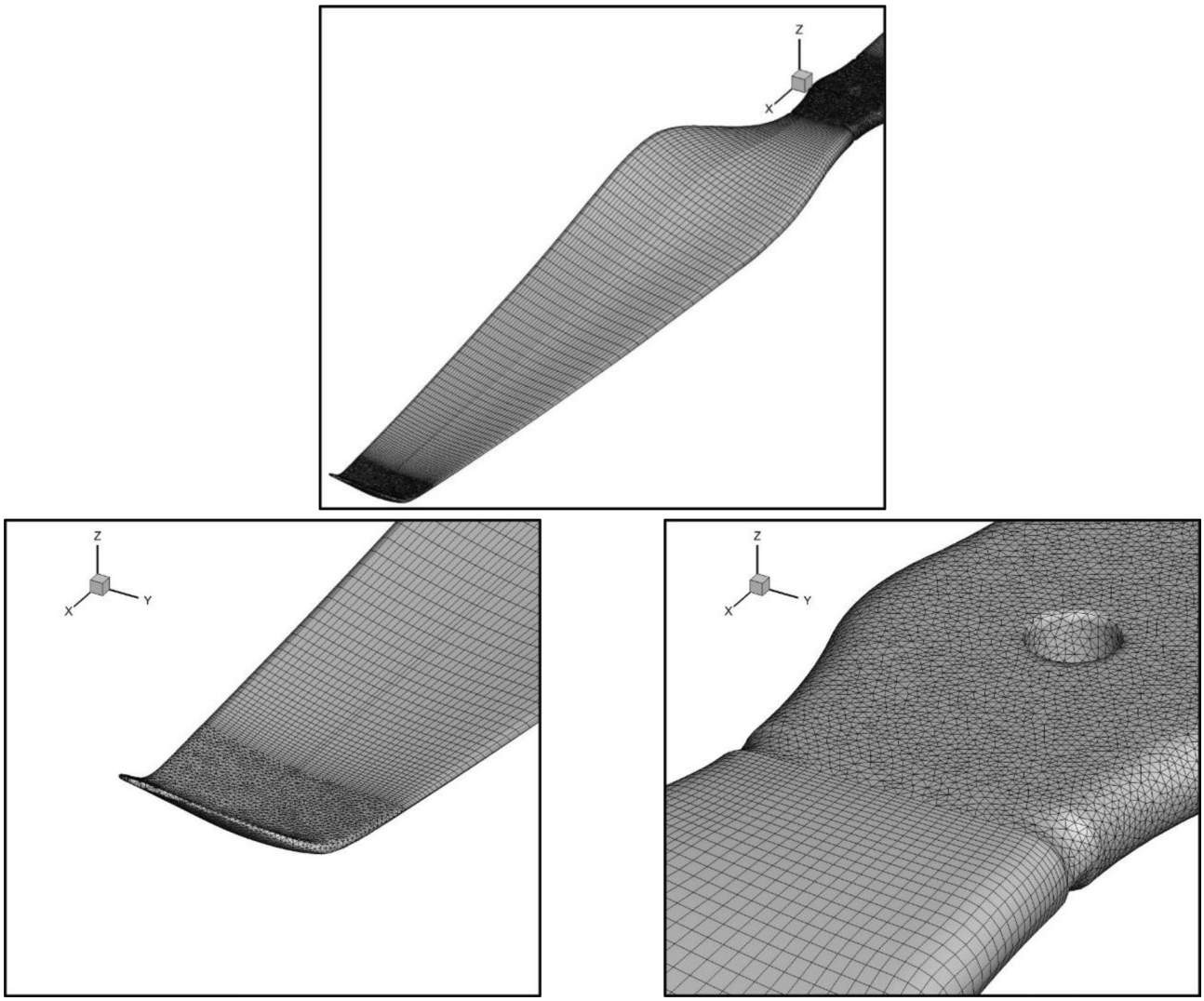


Figure 4. Computational grid generated on the blade surface.

is provided. In Table 4, the thrust and torques for cycle 6 to 7 are averaged. As can be seen, the relative error in torque is small for almost all time steps. However, because changes in thrust are very small for the time step of 0.0002 compared to 0.0001 and the error has increased significantly at higher than this time step; the time step of 0.0002 is appropriate and is selected.

Domain independency

We check this issue in two steps to ensure that present results do not depend on the domain size and that the calculation domain is far enough from the blade. In the first step, all parameters of domains were kept constant, and only the size of the rotational section was changed to ensure that this region was considered large enough. Considering that the case with 2111 rpm is selected for examination. The results of rotational domain independence can be seen in Table 5. As can be seen, changes in thrust and torque are very small for Case 3 compared to Case 2, so rotational domain dimension of $1.5 \times 1.5 \times 0.4$ is appropriate and is selected.

In the next step, now that the size of the rotating area is determined. We will examine the independence of the results from the static computing domain. As before, we considered that the case with 2111 rpm is selected for examination. In Table 6, the thrust and torques for different static domain sizes can be observed. As can be seen, the relative error for the static domain dimension of $10 \times 10 \times 20$ is low with respect to the next static domain, so this static domain is suitable.

The value of y^+ was below 1 in all dimensions of the blade surface, and it is limited to a maximum of 0.52, so the distance of the first layer of the grid is also suitably chosen. In Fig. 7, can be seen on the y^+ of the blade surface in terms of the dimensionless length of the blade.

Numerical validation

For the conditions of 30, 60, and 90% throttle, the simulation was performed and compared with the experimental results. The experimental results were obtained from data from the rotor blade manufacturing company, which

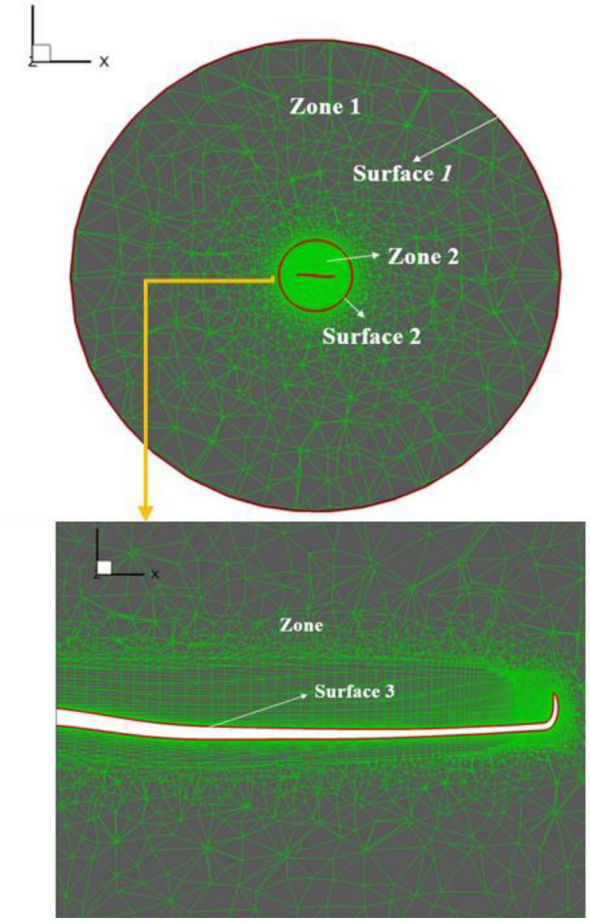


Figure 5. Effective areas in grid.

Cases	Cells Number Surface 1	Cells Number Surface 2	Cells Number Surface 3	Cells Number Zone 1	Cells Number Zone 2	Total Cells Number
Mesh 1	105	564	230,709	20	20	917,105
Mesh 2	164	943	27,909	20	25	1,280,386
Mesh 3	254	2297	32,109	20	30	1,680,119
Mesh 4	949	3926	48,909	40	40	2,630,693

Table 1. The cells per area number and the cells total number for the mesh independency.

Cases	The total number of elements	Thrust (N)	Torque (N.m)	Thrust relative error percentage (%)	Torque relative error percentage (%)
Mesh 1	917,105	104.277	10.0726	–	–
Mesh 2	1,280,386	108.288	9.923	1.507	3.704
Mesh 3	1,680,119	112.33	9.8153	1.097	3.598
Mesh 4	2,630,693	112.787	9.755	0.618	0.405

Table 2. Mesh independency at the angular speed of 896 rpm.

Cases	The total number of elements	Thrust (N)	Torque (N.m)	Thrust relative error percentage (%)	Torque relative error percentage (%)
Mesh 3	1,680,119	707.614	63.653	–	–
Mesh 4	2,630,693	703.508	62.481	0.583	0.516

Table 3. Mesh independency at the angular speed of 2111 rpm.

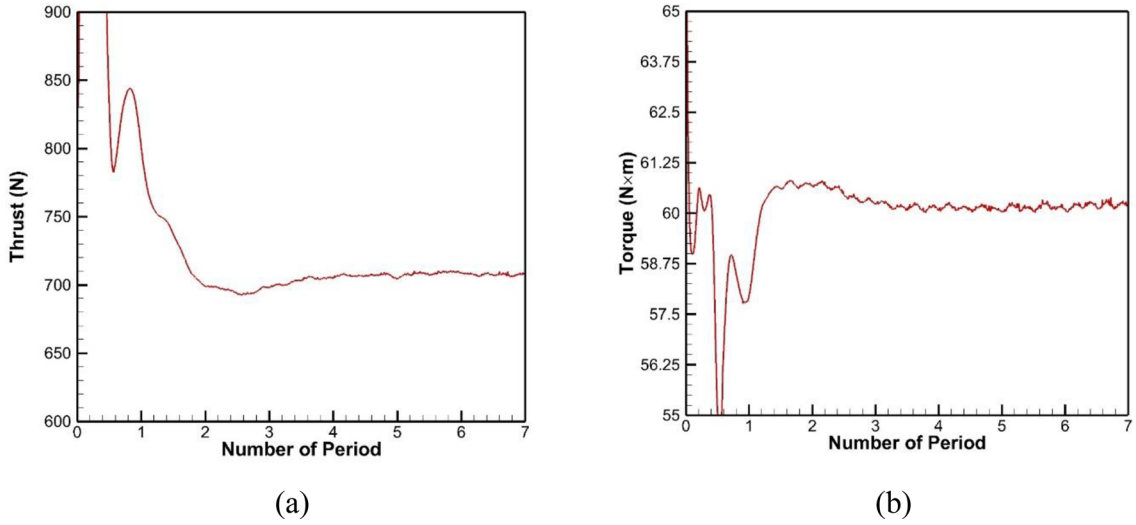


Figure 6. The effect of the number of rotation cycles on the angular speed of 2111 rpm for (**a**) production thrust (**b**) production torque.

Case	Time step	Thrust (N)	Torque (N.m)	Thrust relative error percentage (%)	Torque relative error percentage (%)
Case 1	0.001	658.259	60.382	–	–
Case 2	0.0004	695.201	60.081	5.313	0.501
Case 3	0.0002	707.601	60.198	1.754	0.194
Case 4	0.0001	710.491	60.082	0.404	0.194

Table 4. Time step independency at the angular speed of 2111 rpm.

Case	Rotational domain Dimensions (m)	Thrust (N)	Torque (N.m)	Thrust relative error percentage (%)	Torque relative error percentage (%)
Case 1	0.8 × 0.8 × 0.2	750.394	61.308	–	–
Case 2	1.5 × 1.5 × 0.4	707.601	60.198	5.662	1.81
Case 3	3 × 3 × 0.8	705.348	60.102	0.318	0.159

Table 5. Rotational domain independence at the angular speed of 2111 rpm.

Case	Static domain Dimensions (m)	Thrust (N)	Torque (N.m)	Thrust relative error percentage (%)	Torque relative error percentage (%)
Case 1	5 × 5 × 10	730.694	60.978	–	–
Case 2	10 × 10 × 20	707.601	60.198	3.161	1.279
Case 3	20 × 20 × 40	706.251	60.092	0.191	0.176

Table 6. Static domain independence at the angular speed of 2111 rpm.

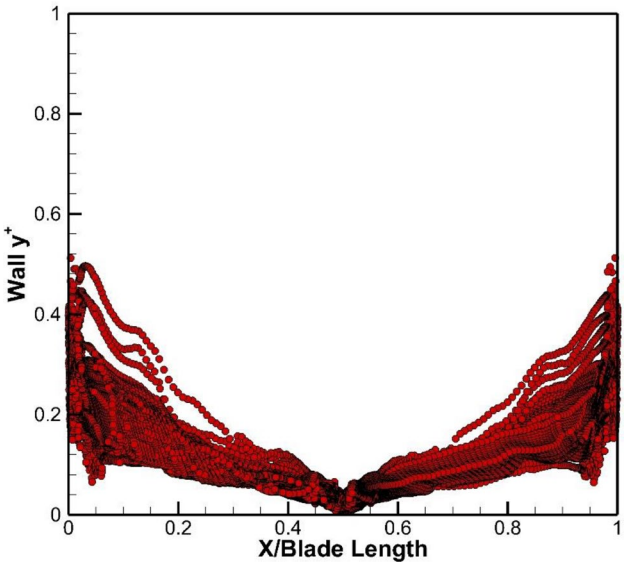


Figure 7. The y^+ of the blade surface in terms of the dimensionless blade length.

is available on the manufacturer's website[39]. The percentage error of the simulation with the experimental results can be seen in Fig. 8. As can be seen, the accuracy of the predicted torque is acceptable. Its error is below 10% and has captured the experimental curve's trend well. In the thrust force chart, the error of the simulation is limited to 17%. The percentage error increases with the increase in rotational speed, which is expected because as the rotational speed increases, the interaction of the rotating area with the still air increases, making the physics of the flow more complicated. When the physics of the flow becomes complicated and the interaction of eddies increases, the turbulence model cannot predict the flow's behavior correctly. The damping effect of the model increases, causing the simulation results to be lower than the real ones, and the error increases. But pay attention, this turbulence model is still more accurate than other models[11] and has fewer errors than other Reynolds-Averaged Navier–Stokes (RANS) models. In other studies, this amount of error has been common for simulations. For example, consider the recent study of Wang et al.[40], who investigated a vertical axis wind turbine with a low rotational speed of 68.78 rpm. They investigated the effect of the wind velocity. According to their results, when the wind speed was low and the interaction between the rotating area and the still air was less, the error was about 10% with the experimental results, and when the speed was high, their error increased to 20%, so their maximum error was 20%. This high error occurred when the vortex structure increased and they had more complex flow physics (similar to the present study). Therefore, our result is similar to previously reported numerical data at the quantitative levels and matches the experimental result. In other words, the maximum error for complex samples is about 20% in other studies. Hence, the accuracy of the simulation is appropriate for low and high rotational speed and the simulation is done correctly.

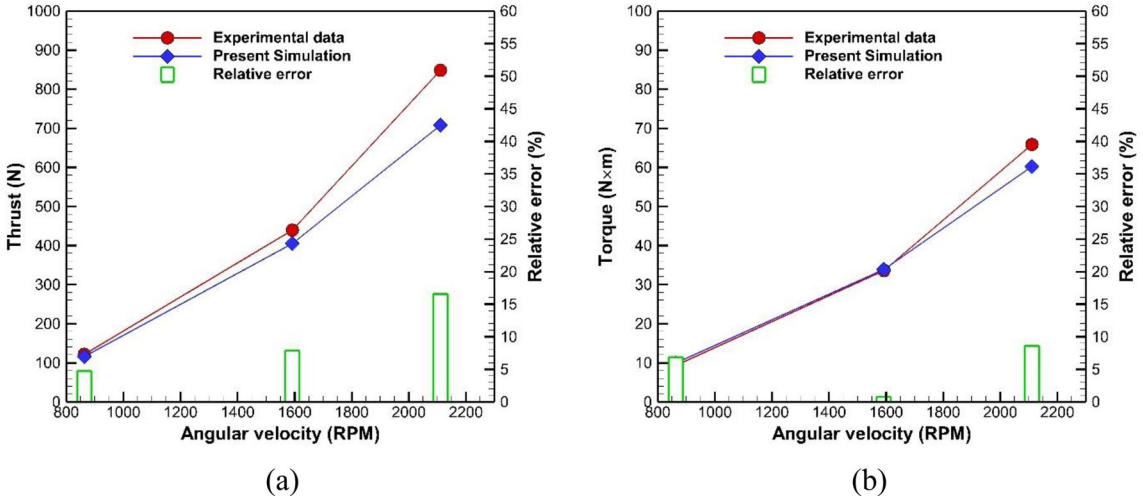


Figure 8. The effect of rotational speed and comparison with experimental results in (**a**) production thrust (**b**) production torque.

Ethical and informed consent

Additional informed consent was obtained from all individual participants for whom identifying information is included in this article.

Results and discussion

Wind inflow effects on the performance of the turbine blade

The angle and direction of the wind's interaction with the objects cause changes in the systems' efficiency[41–43]. In different vehicles, the wind can hit the blade horizontally or vertically. For example, vertical wind with an angle is often observed in wind turbines or propeller motors. The horizontal wind is seen in devices such as quadrotors. On the other hand, in air vehicles with tilt-rotor change mechanisms, we see changes in the wind direction from horizontal to vertical and vice versa. Therefore, all these situations are considered, and the wind direction and angle effect are investigated in this study. In Section "The effect of wind direction ", the effect of wind direction on blade performance was investigated. Then, in Section "Effect of speed in lateral wind", the effect of speed in lateral wind was investigated and finally, the wind angle was examined on the blade's performance. A total of 16 different cases were simulated, and the effect of wind on different conditions was investigated.

The effect of wind direction

In this section, we examined how wind direction affects blade performance. Wind turbines and quadrotors are examples of practical applications where side wind influences their blade efficiency and performance. The blade performance can improve or deteriorate depending on wind direction, angle and amplitude. Generally, wind can act on the blade in two ways: laterally (perpendicular to blade rotation direction) and longitudinally (along blade rotation direction), which can be either in the same direction or opposite to rotation. In the first scenario, we set wind speed to 5 m/s and considered three cases: lateral wind ($Case_{PAR}$) , longitudinal wind in the same direction as rotation ($Case_{DAR}$) and longitudinal wind in the opposite direction of rotation ($Case_{ODAR}$). Figure 9 shows a schematic of these three cases.

The manufacturer's website of the KV29 industrial blade[39] gives thrust and torque for throttles of 30, 60, and 90 percent. Based on the engine's power, the rotational speed of these throttles can be measured, which was available on the manufacturer's website[39] and is 896, 1592, and 2111 rpm, respectively. The percentage error has risen as the rotational speed has increased. This is because as the rotational speed increases, the interaction between the rotating area and the still air becomes more pronounced, leading to a more complex flow physics. With the increased complexity in the flow physics and heightened eddy interactions, the turbulence model struggles to predict flow behavior accurately. Since simulation gave the highest accuracy at 30% of throttle, we considered this rotational speed for investigating the blade under different wind conditions. Figure 10 illustrates the flow physics of these three cases. We plotted pressure contour along with velocity vectors in x and z planes at zero level for these three cases. As seen in z plane at zero level, when wind acts laterally ($Case_{PAR}$) , there is more rotational flow, since flow does not enter perpendicularly to the plane. Wind may bend in that direction if it cannot resist flow induced by rotation, which completely happened in this case. According to figures in the x plane at zero level, when wind flow enters the same direction as rotation, its interaction with secondary flow induced by the rotating blade is low. As a result, the rotational region is less visible, but when wind flow enters

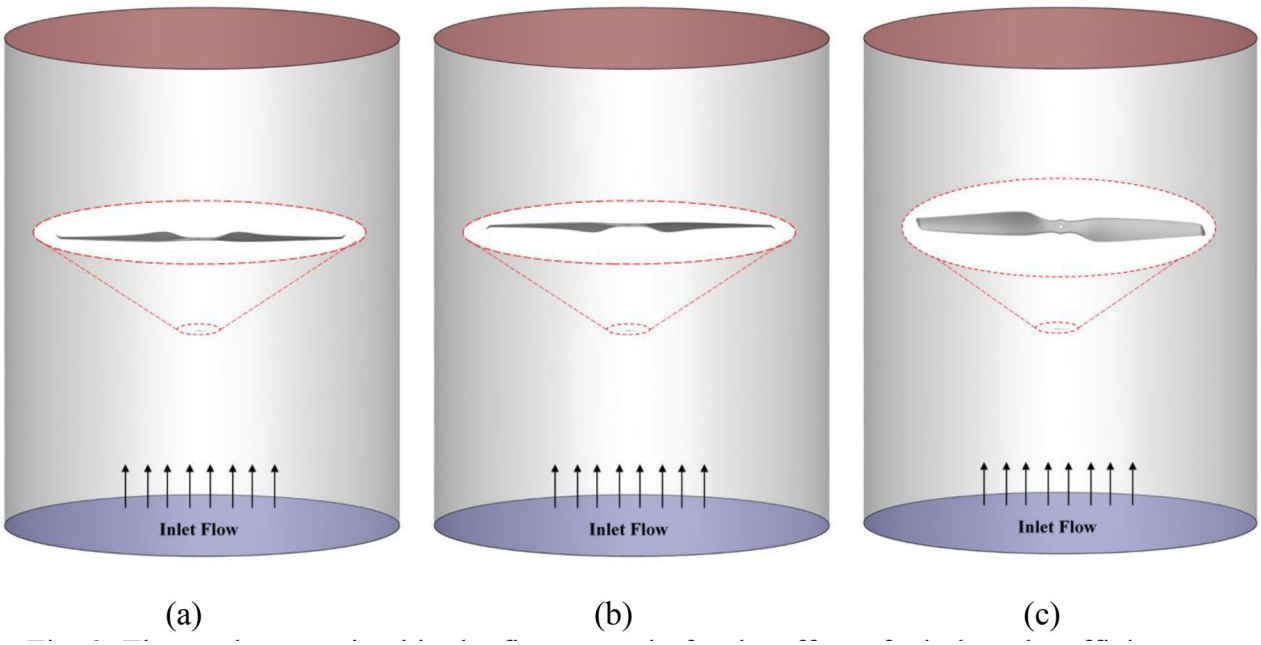


Figure 9. The modes examined in the first scenario for the effect of wind on the efficiency (**a**) $Case_{DAR}$ (**b**)$Case_{ODAR}$ (**c**)$Case_{PAR}$.

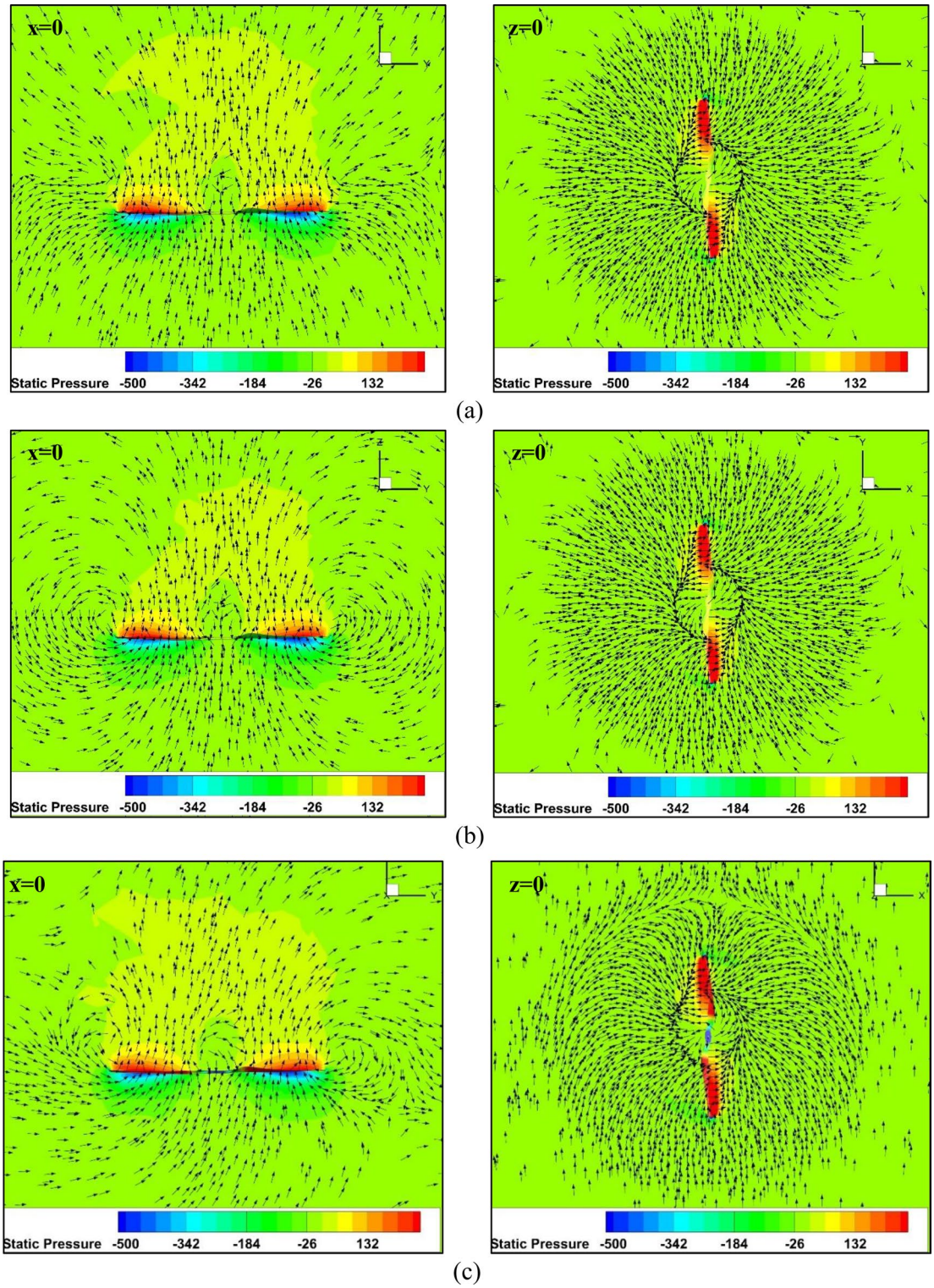


Figure 10. Pressure contour along with velocity vectors in x and z planes equal to zero in (**a**) $Case_{DAR}$ (**b**)$Case_{ODAR}$ (**c**)$Case_{PAR}$.

the opposite direction of rotation, the interaction between wind and secondary flow increases. In $Case_{DAR}$, the flow enters from bottom to top, and in the $Case_{ODAR}$, wind flow is from top to bottom. But in $Case_{PAR}$, the flow enters from left to right; this direction of the wind causes the interaction of the rotating flow with the wind to be different because we see a change in the angle of attack of the entire blade system with the airflow at any moment. But, this sample still does not have enough lateral wind speed to penetrate the secondary flow. On the other hand, by carefully looking at the pressure contour on and behind the blade, it can be stated that there is

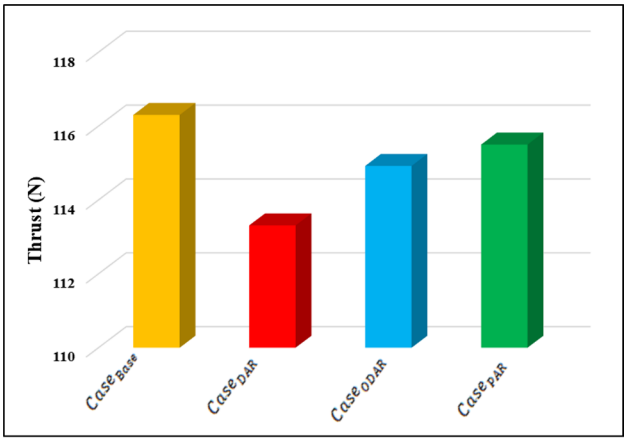
not much change in this speed in the contours, and therefore, the difference in thrust and produced torque due to them will be close to each other.

Figure 11 shows how wind influences the average thrust and torque produced in one cycle compared to case without wind. We can see that when wind speed is too low to overcome secondary flow induced by blade rotation, thrust decreases in all cases (in the same direction, opposite direction, and perpendicular to the direction of rotation). Therefore, in practical operations where we do not encounter high-amplitude winds, wind actually lowers thrust production by 3%, which needs to be considered in calculations. Wind in the same direction as rotation causes most thrust reduction.

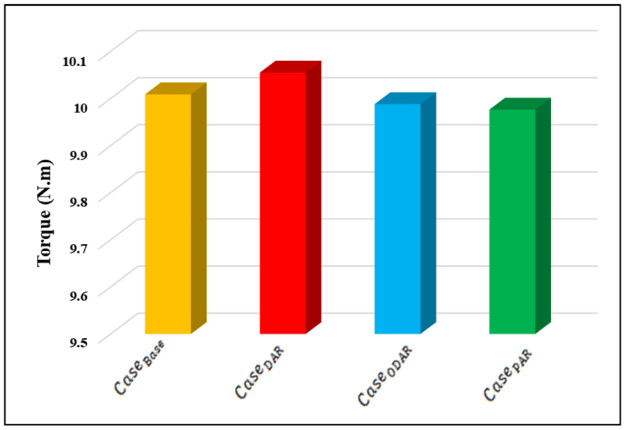
Effect of speed in lateral wind

Figure 12 presents the thrust and torque diagrams in one motion cycle for lateral wind speeds ranging from 0 to 20 m/s in polar and Cartesian coordinates. This mode is typical for quadrotor and vertical wind turbine blades. Four speeds of 5, 10, 15 and 20 m per second were considered for this purpose. The rotational speed was constant at 896 rpm for all simulations. The figure shows that increasing the lateral wind speed resulted in two peaks in the torque diagram, forming a butterfly wing shape in the polar torque diagram and multiple extrema in the torque diagram with increasing speed. However, the thrust diagram had only one peak. A closer inspection reveals that the thrust increased for all angles with increasing speed above 5 m/s, but this was not the case for torque. The maximum thrust shifted slightly to the left with increasing lateral wind speed.

Figure 13 examines the effect of lateral wind speed on the blade flow physics. A snapshot of the rotating blade with the azimuth angle of 315 degrees was considered. The azimuth angle is measured in the horizontal plane between a reference direction (here, x direction) and a blade. In fact, the azimuth angle is the rotation angle of the blade, which starts from the zero angle at the beginning of the rotation. When the blade makes a complete turn and returns to its original position, it becomes 360 degrees. In Fig. 13, the ISO surface for vorticity in the z direction equal to 10 is colored by the magnitude speed value. Since the rotation is in the z direction, the dominant vortex structures are in this direction. The figure indicates that the vortices behind the blade vanished with increasing lateral wind speed and the structures had a higher speed due to the increased wind speed. The lateral



(a)



(b)

Figure 11. Comparison of the average (**a**) thrust (**b**) torque produced in a period of motion for different states of wind application compared to the base state.

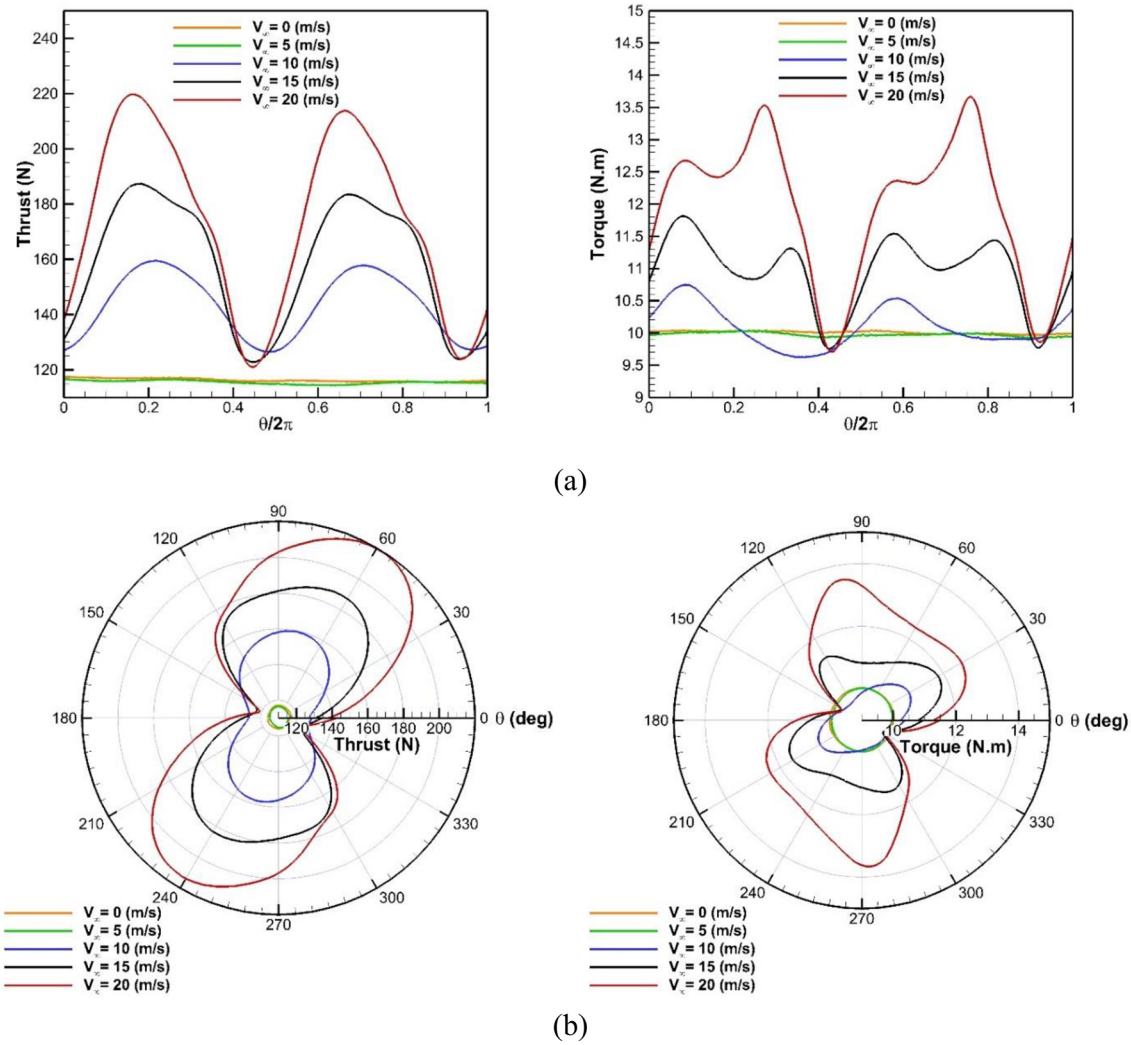


Figure 12. Produced thrust and torque diagram in one cycle of movement for the side wind condition with speeds from 0 to 20 m/s in (**a**) Cartesian (**b**) polar coordinates.

wind inhibited the formation of eddy structures, attenuating the main vortex and enhancing the efficiency and force of thrust.

To derive a formula incorporating the lateral wind's effect, consider Fig. 14. In this figure, the thrust and torque and their dimensionless values are plotted as a function of lateral wind speed compared to the no-wind case. We observed two distinct trends, so we fitted a two-law equation to the values, using a linear graph for lateral wind speeds below 5 m/s, and a parabolic graph for values above 5 m/s. The analytical formula had a very high accuracy. The relationships for thrust and torque along with their accuracy are as follows:

$$\frac{Thrust_{V_\infty}}{Thrust_{V_\infty=0}} = \begin{cases} -0.0014V_\infty + 0.9997;\ R^2 = 0.979,\ V_\infty \le 5m/s \\ -0.0011{V_\infty}^2 + 0.0602V_\infty + 0.7246;\ R^2 = 0.996,\ V_\infty > 5m/s \end{cases} \tag{7}$$

$$\frac{Torque_{V_\infty}}{Torque_{V_\infty=0}} = \begin{cases} -0.0006V_\infty + 0.9998;\ R^2 = 0.974,\ V_\infty \le 5m/s \\ 0.0009{V_\infty}^2 - 0.0083V_\infty + 1.0134;\ R^2 = 0.993,\ V_\infty > 5m/s \end{cases} \tag{8}$$

Effect of wind angle

The wind hit the blade directly in the previous sections. In this section, the wind speed is considered to be 10 m/s, and the wind with some angle hits the blade. Two general modes were considered: the angle was in the direction of the rotation axis or the other longitudinal direction. The angle ψ was the angle between the direct side wind and the axis of rotation, and the angle ϕ was the angle of the side wind with the other direction of the blade, as shown in Fig. 15.

The effect of these two angles was discussed independently to simplify the aerodynamic analysis and reduce the physical complexity. The effect of changes in ϕ angle assuming $\psi = 0$ and then the effect of changes in ψ angle assuming $\phi = 0$ on flow physics, thrust, and torque was investigated. Figure 16 shows the influence of

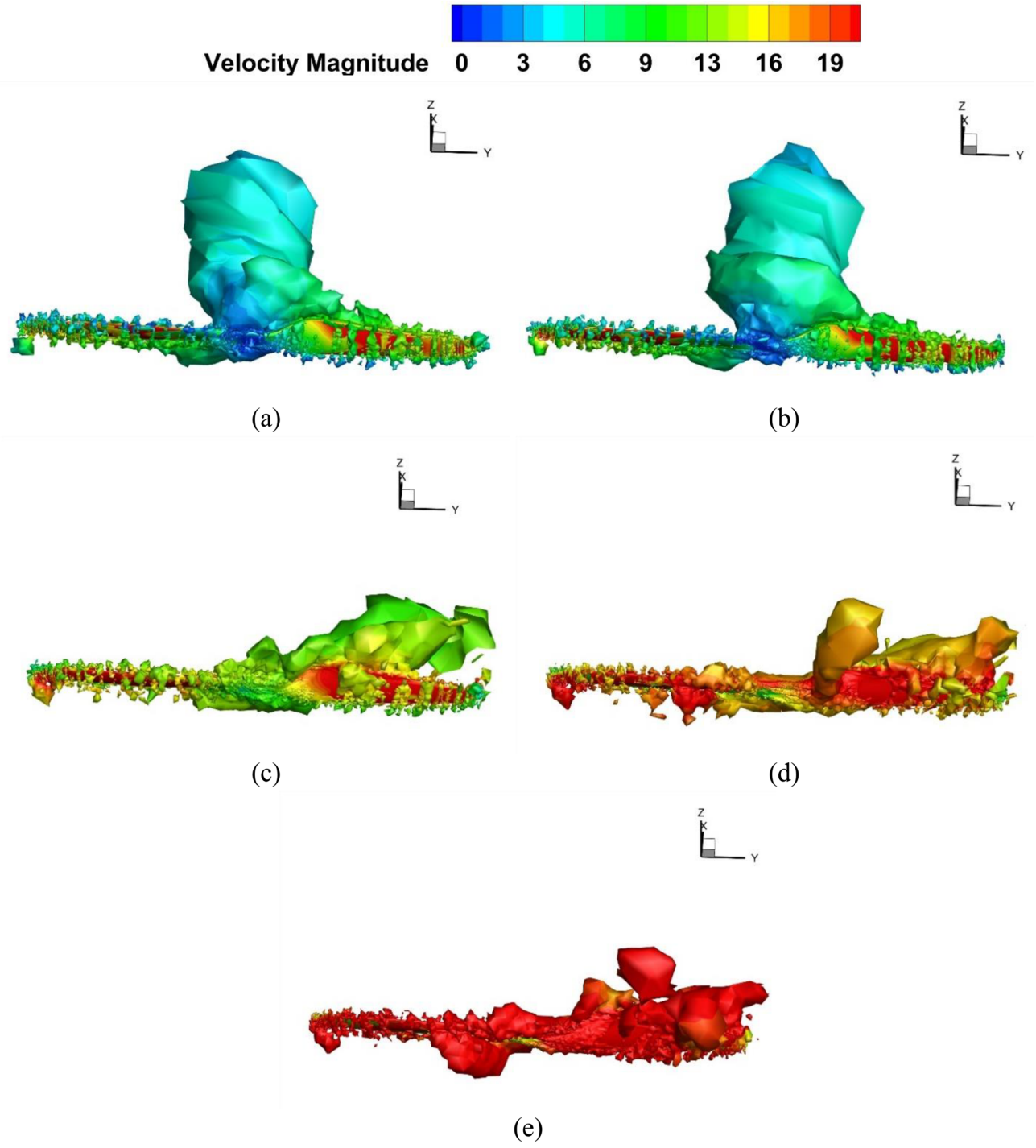


Figure 13. The ISO level for vorticity in the z direction is 10, colored with the velocity magnitude for the rotational speed of 896 rpm with a side wind of (**a**) 0, (**b**) 5, (**c**) 10, (**d**) 15 and (**e**) 20 m/s.

ϕ angle on the thrust and torque diagrams in one motion cycle. According to the figure, ϕ angle only caused a phase difference in the motion and did not affect the thrust amplitude and torque. Increasing the angle resulted in a regular process of changes, which led to more phase advances in the graphs. The influence of the angle in the polar diagram was also evident. It caused the oval shape, which was due to the presence of lateral wind, to have a rotational relative to the origin, and this caused different thrusts to be produced at different times. However, since the general shape did not change, the average thrust did not change either.

A point of motion was marked in Fig. 16 under part (b) to examine the flow physics further. Figure 17 shows the influence of ϕ angle on the ISO surface for Q criterion equal to 10, colored by velocity magnitude for rotational speed of 896 rpm with wind of 10 m/s. The blade angle was 320 degrees for this case. According to the figure, vortex and eddy structures were also stretched in this direction in the case of wind with zero angles since the wind entered from top to bottom. As ϕ angle increased, these structures extended along the x-axis and

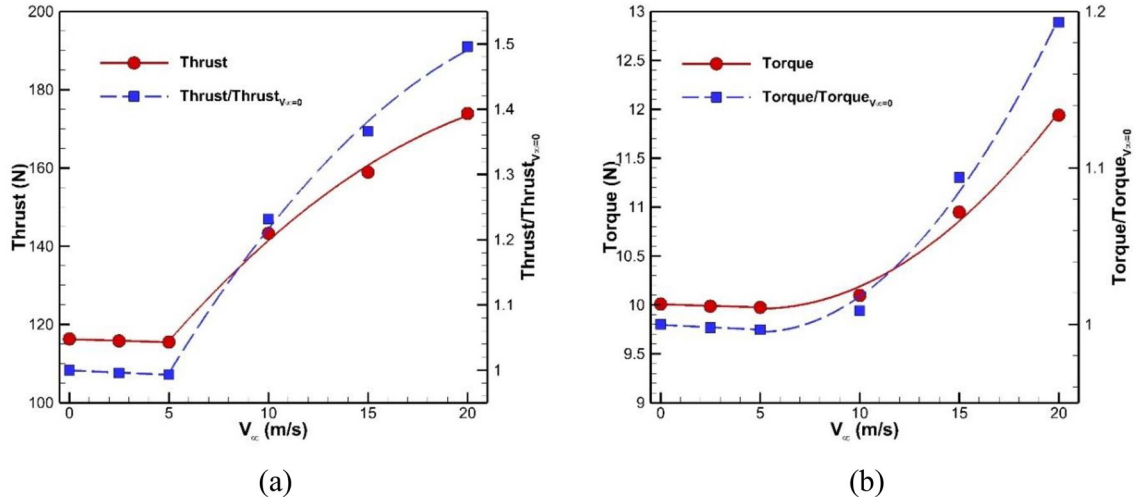


Figure 14. The effect of the side wind speed compared to the no wind condition for the angular speed of 896 rpm on mean of (**a**) thrust (**b**) torque production during one period of blade motion.

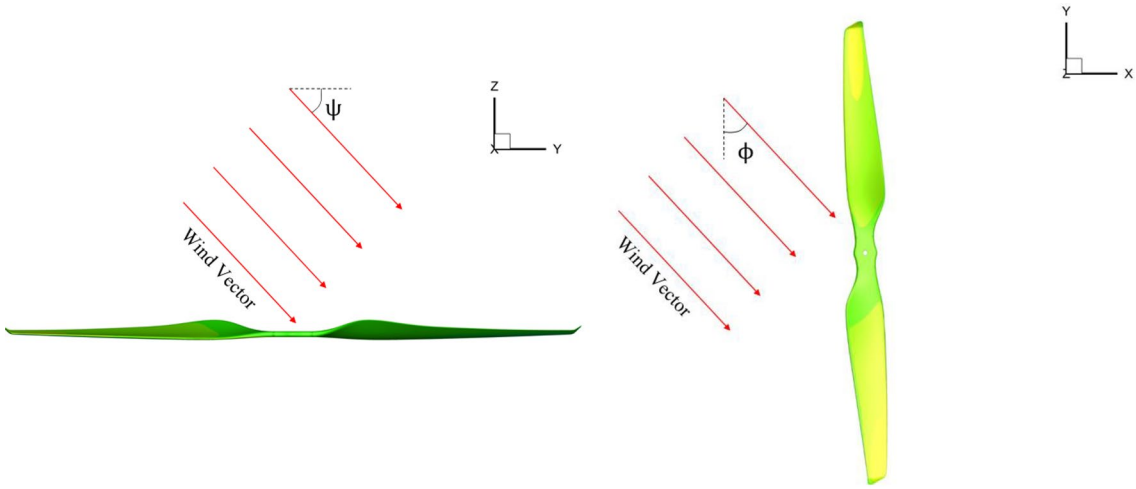


Figure 15. Definition of wind angles.

expanded. This expansion caused an asymmetric structure of vortices with respect to the y-axis, and as the angle increased, this asymmetry also increased.

The effect of changing ϕ angle can be seen in the pressure coefficient diagram with the wind of 10 m/s at Sects. 0.2, 0.4 and 0.6 m at blade angle of 320 degrees in Fig. 18. According to this figure, thrust production at 90 degrees should be higher than other angles because pressure difference between the upper and lower surface of blade in these three sections for ϕ angle was always higher than others. Of course, it should be noted that the general trend of thrust with these three cross sections is not completely clear because in integral thrust production, pressure difference for the entire length of the blade and infinite parts was considered, and the general trend cannot be fully predicted with 3 cases. In these three considered sections, with an increase in wind angle, pressure difference also increased due to an increase in the angle of attack that the blade section faced with respect to flow.

Following the effect of wind angle, the effect of ψ angle was investigated. Figure 19 presents the thrust and torque diagrams for different ψ angles in Cartesian and polar coordinates in one motion cycle. The figure shows that ψ angle damped the oscillatory motion in the blade motion. The amplitude of motion decreased and the graph became smoother as the angle increased. An almost straight line similar to no side wind at 60 and 90 degrees was observed, which was attributed to the lack of change in the effective angle of attack during the rotational cycle. The component against the direction of rotation increased as ψ angle increased, which, as investigated in previous sections, caused a greater reduction in thrust compared to direct side wind. Some small fluctuations are seen in higher angles due to the presence of small eddies and confused flow. The figure shows that increasing this angle from zero decreased the blade's overall thrust because wind flow was opposite to the direction of rotation. The wind had enough momentum due to its high speed, which strongly resisted thrust production and greatly reduced thrust force. A point of motion during a cycle was considered to investigate flow physics further. The considered point can be seen in Fig. 19 (b).

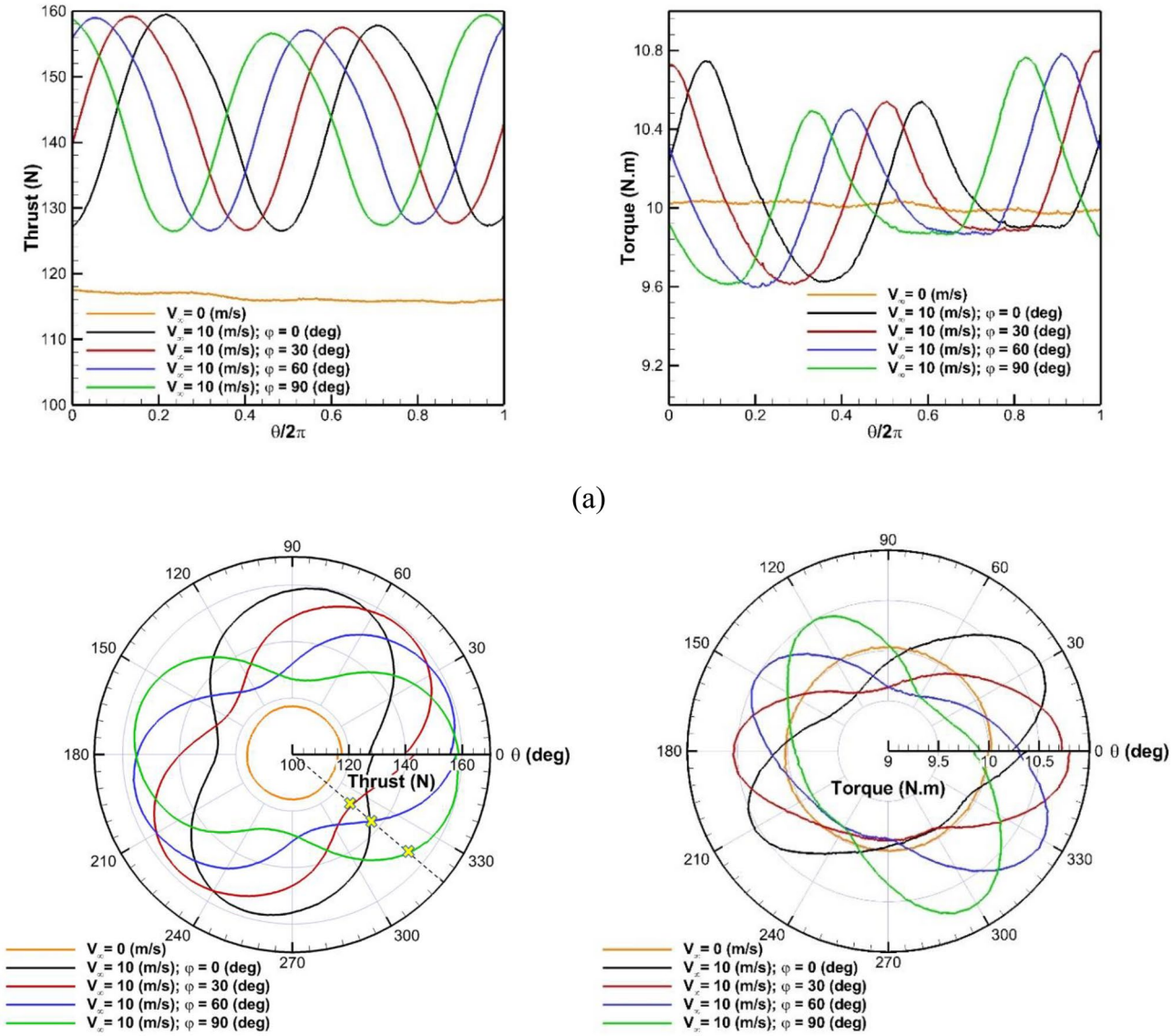


Figure 16. Thrust diagram and production torque in one cycle of movement for wind condition with 10 m/s with different angles ϕ in (**a**) Cartesian (**b**) polar coordinates.

Figure 20 illustrates the effect of ψ angle on ISO surface for Q criterion equal to 10, colored by velocity magnitude for rotational speed of 896 rpm with wind of 10 m/s at blade angle of 320 degrees, which is specified. The figure indicates that rotational structures induced by blade rotation were greatly reduced with an increase of ψ angle. The detachment was visible behind the blade in all cases with increased angle. However, main and rotational structures that resulted from rotation and thrust production were not visible with increasing angle. On the other hand, the remaining structures also had lower speed than when ψ was equal to zero, which was the reason for this decrease in wind resistance against the rotation of airflow and its neutralization. On the other hand, rotational structures decreased, but symmetry along the x and y axis increased as the angle increased.

The effect of changing ψ angle can be seen in the pressure coefficient diagram with wind of 10 m/s at Sects. 0.2, 0.4 and 0.6 m at a blade angle of 320 degrees in Fig. 21. The figure shows that the increase in ψ angle caused a sharp decrease in the pressure difference between top and bottom of the blade in all three cross-sections, and this caused thrust in non-zero ψ to be even less than in case of no wind. The pressure coefficient diagram in two cross sections of 0.4 and 0.6 had a regular trend with increasing angles. It indicated that flow physics in these two cross sections did not change much due to the change of angle. However, in the cross section of 0.2, the graphs were completely different and indicated different physics and formation of different vortices in these areas due to the increase of ψ angle.

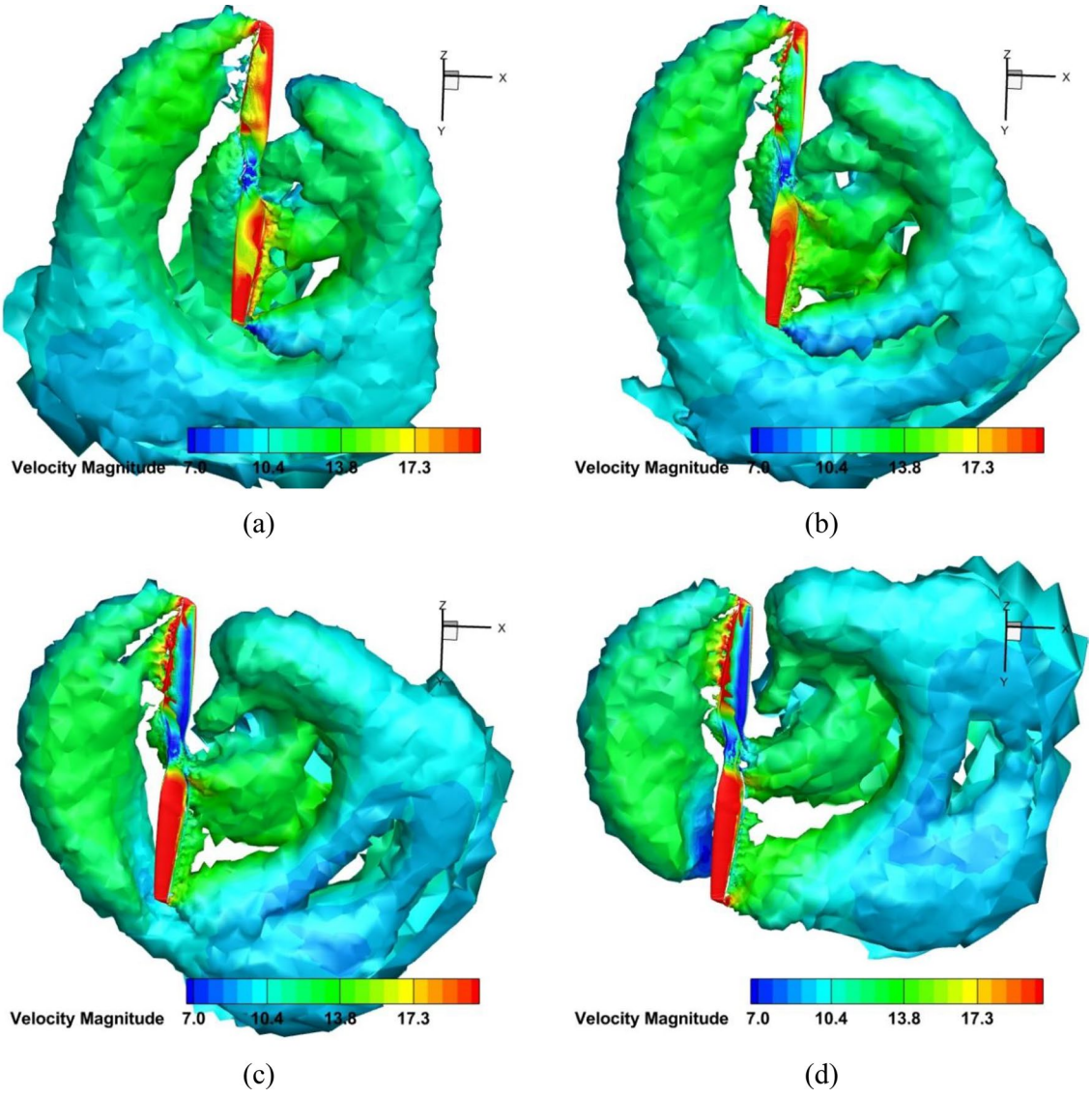


Figure 17. The ISO surface for the Q criterion equals 10 is colored with the velocity magnitude for the rotational speed of 896 rpm with a wind of 10 m/s for the angles $\psi = 0$ and (**a**) $\phi = 0$ (**b**) $\phi = 30$ (**c**) $\phi = 60$ (**d**) $\phi = 90$ degrees.

Dimensionless thrust and torque were plotted for different angles ϕ so that the effect of this angle could be seen better. The same was done for ψ angle. In Fig. 22, the effect of these wind angles on thrust and torque compared to no-angle mode at a rotational speed of 896 rpm was investigated. According to the figure, changes in ϕ angle did not affect torque and thrust produced in a cycle, and it was a constant value at each angle, but thrust and torque produced decreased parabolically with the change of ψ angle.

Therefore, we have for the angle ϕ:

$$\frac{Thrust_{\phi}}{Thrust_{\phi=0}} = 1 \tag{9}$$

$$\frac{\boldsymbol{Torque_{\phi}}}{\boldsymbol{Torque_{\phi=0}}} = 1 \tag{10}$$

And we have for the angle ψ:

$$\frac{Thrust_{\psi}}{Thrust_{\psi=0}} = 5 \times 10^{-5}\psi^2 - 0.0132\psi + 1.0091;\ R^2 = 0.9951 \tag{11}$$

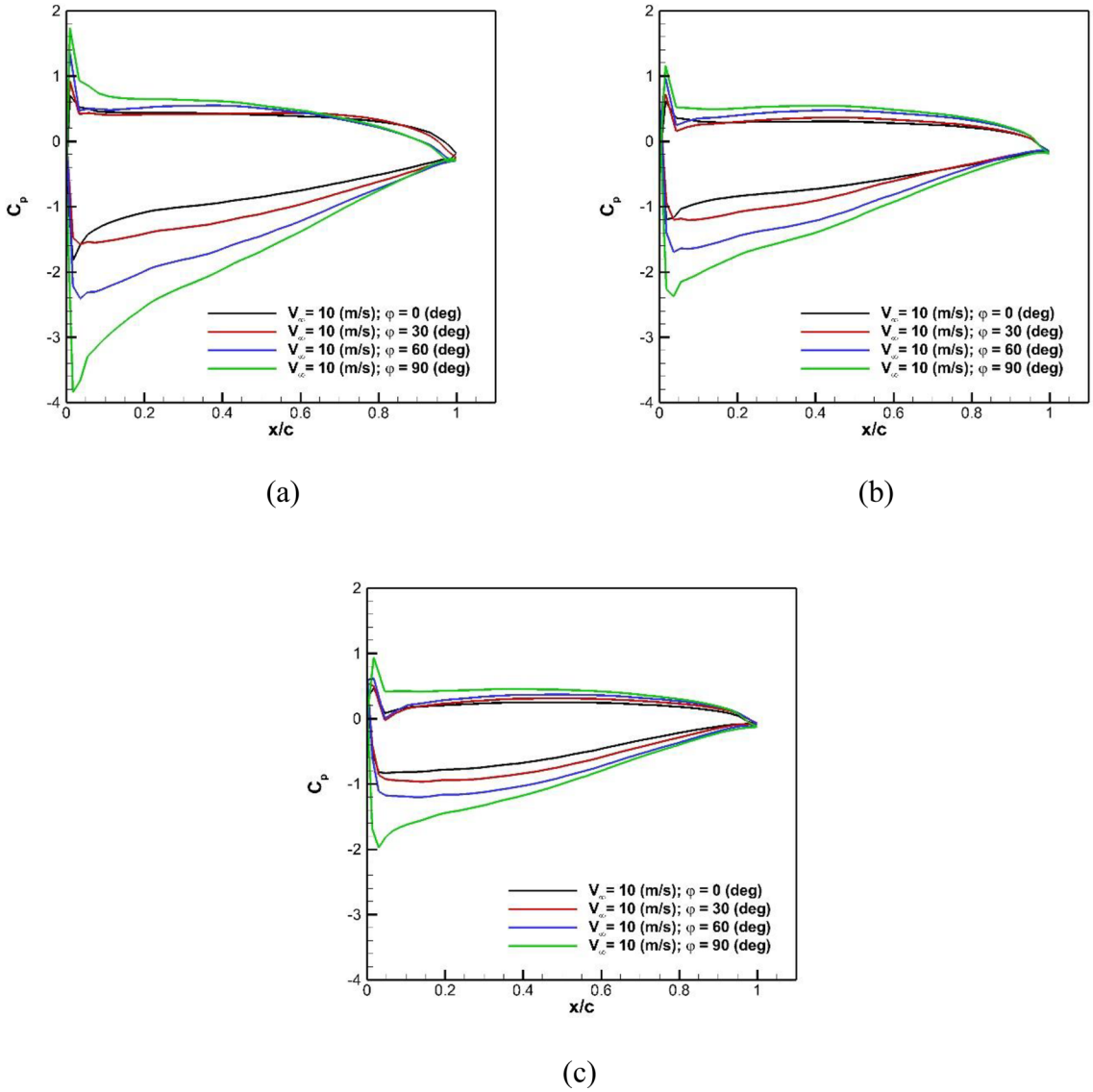


Figure 18. The influence of the ϕ angle on the pressure coefficient diagram in the presence of a wind of 10 m/s on the surface of sections (**a**) 0.2 (**b**) 0.4 (**c**) 0.6 m.

$$\frac{Torque_{\psi}}{Torque_{\psi=0}} = -3 \times 10^{-5}\psi^2 - 0.0003\psi + 1.0136;\ R^2 = 0.9351 \tag{12}$$

Conclusions

This paper presents a CFD numerical study of the unsteady aerodynamic performance of a rotor blade under different wind conditions. The rotor blade used is the U15XXL Combo KV29 industrial blade, which has no available CAD model. Therefore, a 3D laser scanning technique was applied to generate the CAD model from a mesh file based on an experimental sample. A 3D transient CFD simulation using the dynamic mesh method was performed to analyze the rotor dynamics. The URANS equations with the k-ω SST turbulence model were solved to simulate the rotor with angular velocities of 863, 1592 and 2111 rpm under various wind directions, speeds and angles. The main findings of this study are summarized as follows.

(1) The study reveals that when the wind speed is insufficient to counter the secondary flow induced by the blade rotation, the thrust decreases across all wind directions. This finding has significant practical implications, particularly in scenarios where high-amplitude winds are infrequent. In such cases, the wind can actually reduce the thrust production by 3%, a factor that should be considered in calculations and operational planning.

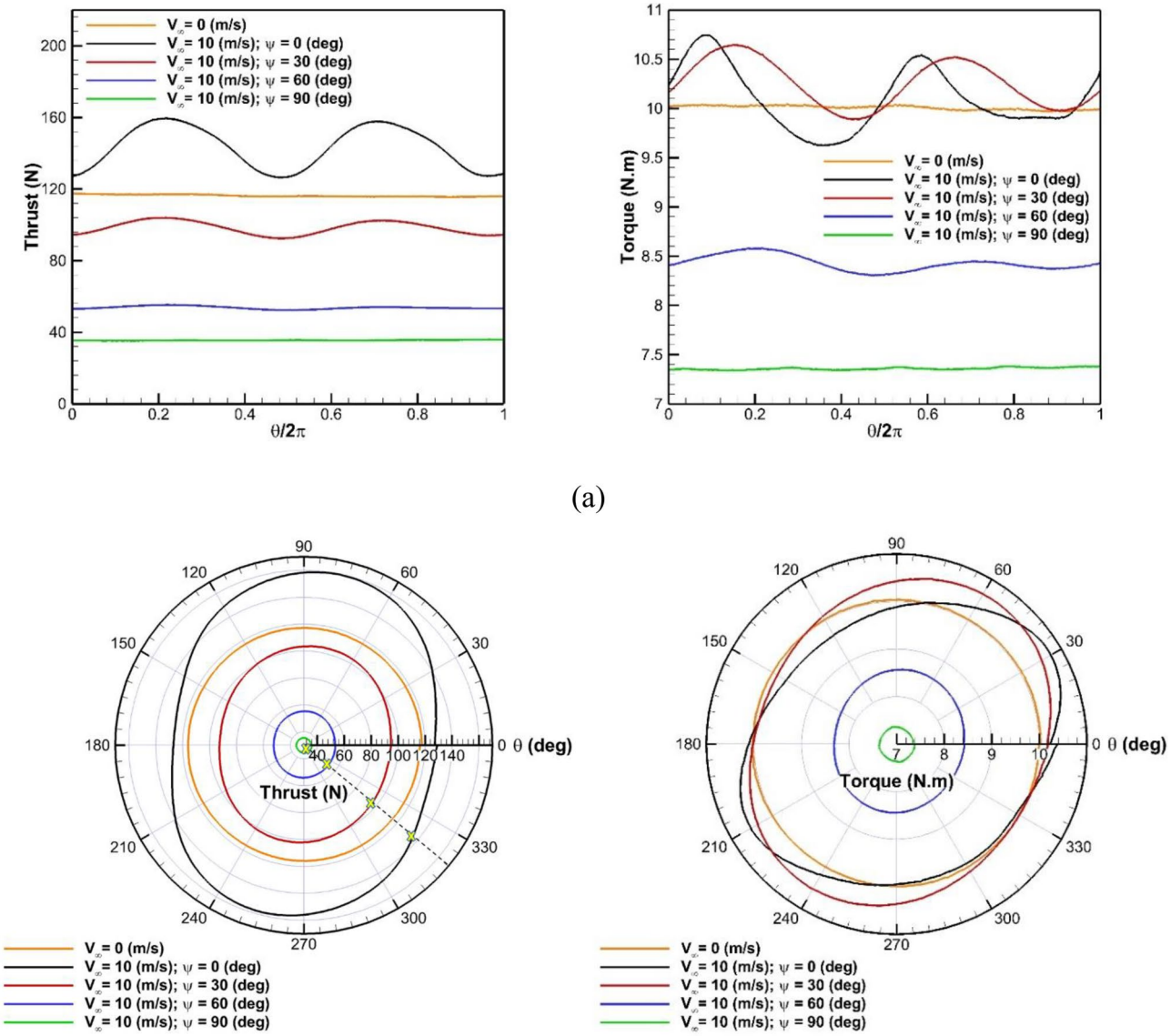


Figure 19. Thrust diagram and produced torque in one revolution for wind condition with 10 m/s with different angles ψ in (**a**) Cartesian (**b**) polar coordinates.

(2) The study demonstrates that lateral wind, a more common occurrence in quadrotor applications, has a less detrimental effect on thrust production compared to opposite wind. This comparative advantage of lateral wind provides reassurance and confidence in the performance of quadrotors in such conditions.

(3) Increasing the lateral wind speed results in two peaks in the torque curve, forming a butterfly wing shape in the polar torque plot and multiple extrema in the torque curve with increasing speed. However, the thrust curve has only one peak. A closer inspection reveals that the thrust increases for all angles with increasing speed above 5 m/s, but this is not the case for torque. The maximum thrust shifts slightly to the left with increasing lateral wind speed.

(4) The ϕ angle only causes a phase difference in the motion and does not affect the thrust amplitude and torque. Increasing the angle results in a regular process of changes, leading to more phase advances in the curves.

(5) The ψ angle dampens the oscillatory motion in the blade motion. The motion amplitude decreases and the curve becomes smoother as the angle increases. A nearly straight line similar to no side wind is observed at 60 and 90 degrees, which is attributed to the constant effective angle of attack during the rotation cycle. The component opposite to the rotation direction increases as ψ angle increases, which causes a greater reduction in thrust compared to direct side wind.

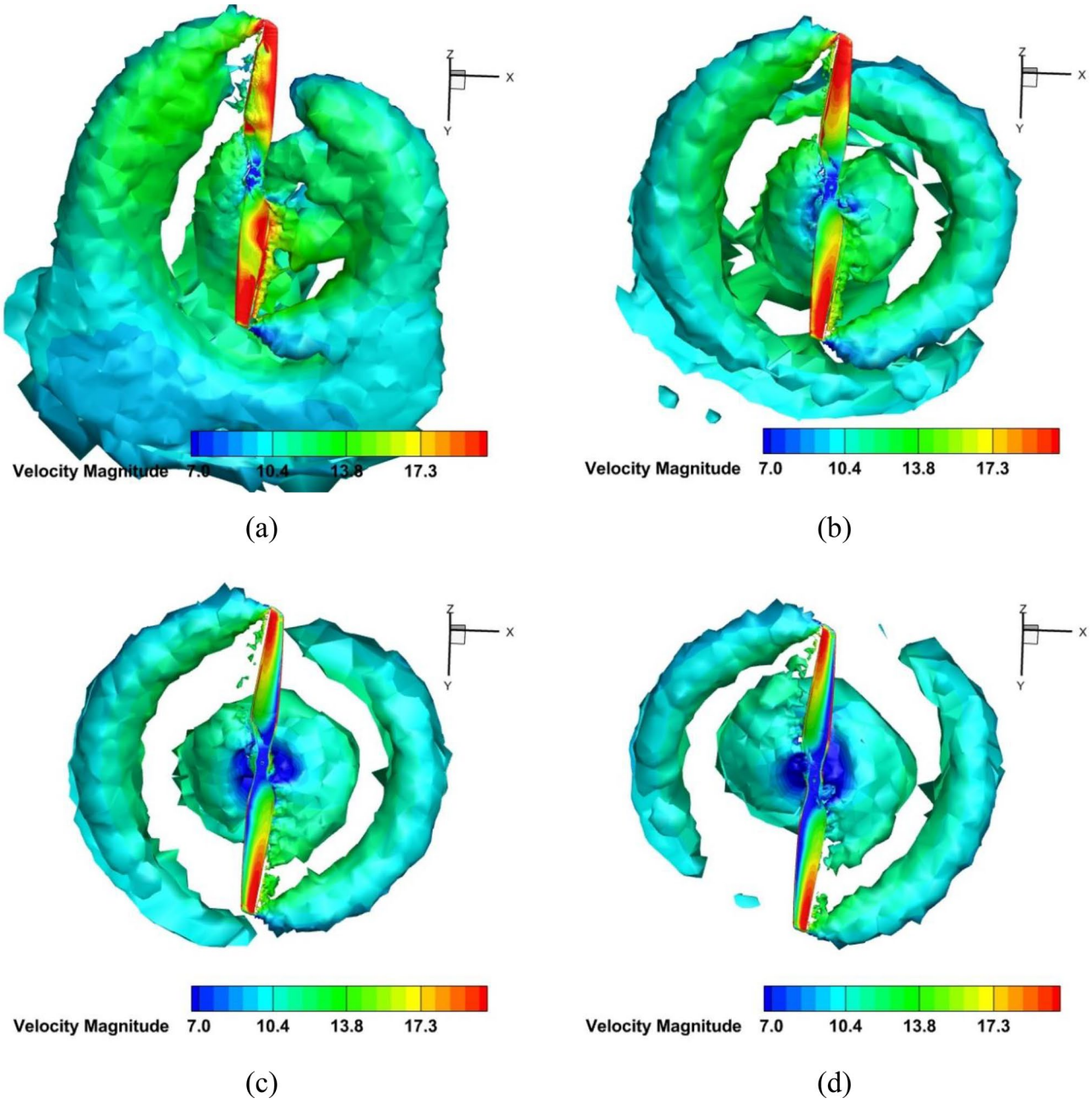


Figure 20. ISO surface for Q criterion equal to 10 colored with velocity magnitude for rotational speed 896 rpm with wind 10 m/s for angles $\phi = 0$ and (**a**) $\psi = 0$ (**b**)$\psi = 30$ (**c**) $\psi = 60$ (**d**) $\psi = 90$ degrees.

(6) In the case of wind with zero angles, since wind enters from top to bottom, vortex and eddy structures are also stretched in this direction, and as ϕ angle increases, these structures extend along the x-axis and expand. This expansion causes an asymmetric structure of vortices with respect to the y-axis, and as the angle increases, this asymmetry also increases. The rotational structures induced by blade rotation are greatly reduced with an increase of ψ angle. The detachment is visible behind the blade in all cases with increased angle, but main and rotational structures that resulted from rotation and thrust production are not visible with increasing angle.

(7) The study provides analytical expressions for estimating thrust and torque based on each investigated parameter. These expressions are designed to be practical and user-friendly, facilitating the work of other researchers who may need to use them for their own analysis.

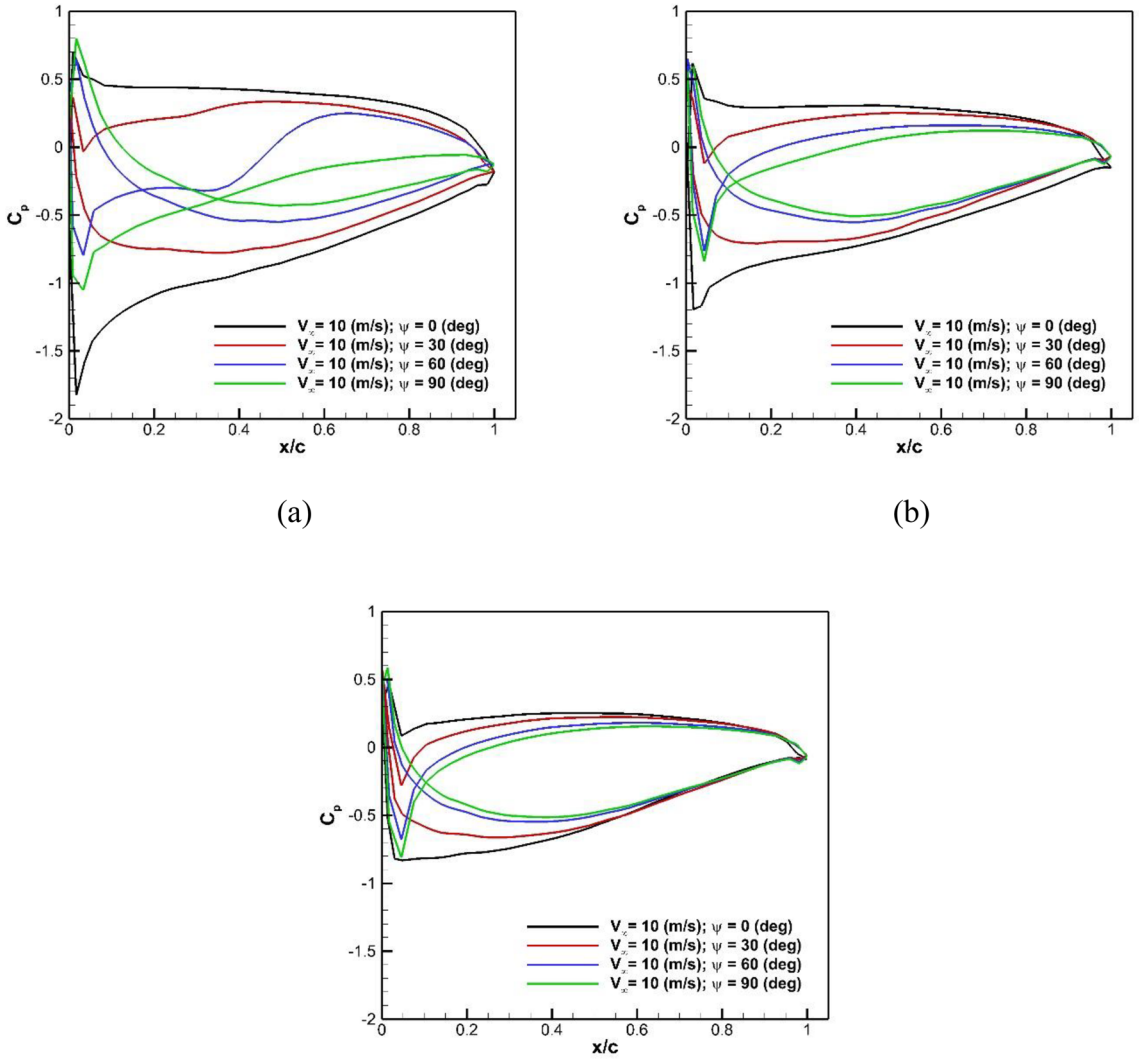


Figure 21. The influence of the ψ angle on the pressure coefficient diagram in the presence of a wind of 10 m/s on the surface of sections (**a**) 0.2 (**b**) 0.4 (**c**) 0.6 m.

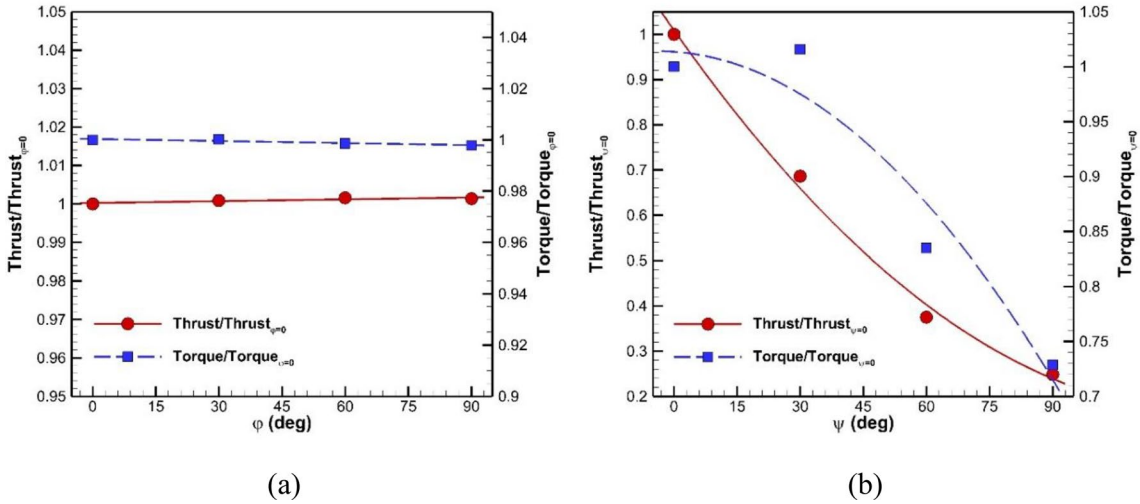


Figure 22. The ratio of thrust and produced torque to the no-angle mode at the angular speed of 896 rpm and wind of 10 m/s with different angles (**a**) ϕ (**b**) ψ.

Data availability

Data will be made available on request. If someone wants to request the data from this study, they can contact corresponding author (Si. Kasmaiee).

Received: 23 April 2024; Accepted: 14 August 2024
Published online: 19 August 2024

References

1. Tadjfar, M., Kasmaiee, S. & Noori, S. Continuous blowing jet flow control optimization in dynamic stall of NACA0012 airfoil. in Fluids Engineering Division Summer Meeting vol. 83723 V002T03A029 (2020). https://doi.org/10.1115/FEDSM2020-20147
2. Ascendance Flight Technologies. ATEA https://www.ascendance-ft.com/ (2021) https://www.ascendance-ft.com/.
3. Tadjfar, M., Kasmaiee, S. & Noori, S. Optimization of NACA 0012 airfoil performance in dynamics stall using continuous suction jet. in *Fluids Engineering Division Summer Meeting vol. 83723* V002T03A028 (2020). https://doi.org/10.1115/FEDSM2020-20147
4. Bangura, M., Mahony, R. & others. Nonlinear dynamic modeling for high performance control of a quadrotor. (2012).
5. Chiang, W.-C., Li, Y., Shang, J. & Urban, T. L. Impact of drone delivery on sustainability and cost: Realizing the UAV potential through vehicle routing optimization. *Appl. Energy* **242**, 1164–1175. https://doi.org/10.1016/j.apenergy.2019.03.117 (2019).
6. Estrada, M. A. R. & Ndoma, A. The uses of unmanned aerial vehicles–UAV's-(or drones) in social logistic: Natural disasters response and humanitarian relief aid. *Procedia Comput. Sci.* **149**, 375–383. https://doi.org/10.1016/j.procs.2019.01.151 (2019).
7. Paz, C., Suárez, E., Gil, C. & Vence, J. Assessment of the methodology for the CFD simulation of the flight of a quadcopter UAV. *J. Wind Eng. Ind. Aerodyn.* **218**, 104776. https://doi.org/10.1016/j.jweia.2021.104776 (2021).
8. Hsiao, F.-B., Bai, C.-J. & Chong, W.-T. The performance test of three different horizontal axis wind turbine (HAWT) blade shapes using experimental and numerical methods. *Energies* **6**, 2784–2803. https://doi.org/10.3390/en6062784 (2013).
9. Brocklehurst, A. & Barakos, G. N. A review of helicopter rotor blade tip shapes. *Prog. Aerosp. Sci.* **56**, 35–74. https://doi.org/10.1016/j.paerosci.2012.06.003 (2013).
10. Barakos, G. N. & Garcia, A. J. CFD analysis of hover performance of rotors at full-and model-scale conditions. *Aeronaut. J.* **120**, 1386–1424. https://doi.org/10.1017/aer.2016.58 (2016).
11. Ali, Q. S. & Kim, M.-H. Unsteady aerodynamic performance analysis of an airborne wind turbine under load varying conditions at high altitude. *Energy Convers. Manag.* **210**, 112696. https://doi.org/10.1016/j.enconman.2020.112696 (2020).
12. Lanzafame, R., Mauro, S. & Messina, M. Wind turbine CFD modeling using a correlation-based transitional model. *Renew. Energy* **52**, 31–39. https://doi.org/10.1016/j.renene.2012.10.007 (2013).
13. Bai, C.-J., Chen, P.-W. & Wang, W.-C. Aerodynamic design and analysis of a 10 kW horizontal-axis wind turbine for Tainan. *Taiwan. Clean Technol. Environ. Policy* **18**, 1151–1166. https://doi.org/10.1007/s10098-016-1109-z (2016).
14. Tong, G., Li, Y., Tagawa, K. & Feng, F. Effects of blade airfoil chord length and rotor diameter on aerodynamic performance of straight-bladed vertical axis wind turbines by numerical simulation. *Energy* **265**, 126325. https://doi.org/10.1016/j.energy.2022.126325 (2023).
15. Yoon, J. & Lee, J. Parameter analysis and design for the hovering thrust of a quad-rotor air vehicle using CFD and design of experiment. *J. Mech. Sci. Technol.* **32**, 781–791. https://doi.org/10.1007/s12206-018-0126-0 (2018).
16. Lei, Y. & Wang, H. Aerodynamic performance of a quadrotor MAV considering the horizontal wind. *IEEE Access* **8**, 109421–109428. https://doi.org/10.1109/ACCESS.2020.3002706 (2020).
17. Cai, X., Gu, R., Pan, P. & Zhu, J. Unsteady aerodynamics simulation of a full-scale horizontal axis wind turbine using CFD methodology. *Energy Convers. Manag.* **112**, 146–156. https://doi.org/10.1016/j.enconman.2015.12.084 (2016).
18. Al-Ghriybah, M., Zulkafli, M. F., Didane, D. H. & Mohd, S. The effect of inner blade position on the performance of the Savonius rotor. *Sustain. Energy Technol. Assessments* **36**, 100534. https://doi.org/10.1016/j.seta.2019.100534 (2019).
19. Takahashi, S., Hata, Y., Ohya, Y., Karasudani, T. & Uchida, T. Behavior of the blade tip vortices of a wind turbine equipped with a brimmed-diffuser shroud. *Energies* **5**, 5229–5242. https://doi.org/10.3390/en5125229 (2012).
20. Zhang, Z. *et al.* Numerical investigation of the aerodynamic and wake characteristics of a floating twin-rotor wind turbine under surge motion. *Energy Convers. Manag.* **283**, 116957. https://doi.org/10.1016/j.enconman.2023.116957 (2023).
21. Cai, J. & Gunasekaran, S. Frequency Response of RC Propellers to Streamwise Gusts in Forward Flight. *Wind* **3**, 253–272. https://doi.org/10.3390/wind3020015 (2023).
22. Paz, C., Suárez, E., Gil, C. & Baker, C. CFD analysis of the aerodynamic effects on the stability of the flight of a quadcopter UAV in the proximity of walls and ground. *J. Wind Eng. Ind. Aerodyn.* **206**, 104378. https://doi.org/10.1016/j.jweia.2020.104378 (2020).
23. Tanabe, Y., Sugawara, H., Sunada, S., Yonezawa, K. & Tokutake, H. Quadrotor drone hovering in ground effect. *J. Robot. Mechatronics* **33**, 339–347. https://doi.org/10.20965/jrm.2021.p0339 (2021).
24. Sagharichi, A., Zamani, M. & Ghasemi, A. Effect of solidity on the performance of variable-pitch vertical axis wind turbine. *Energy* **161**, 753–775. https://doi.org/10.1016/j.energy.2018.07.160 (2018).
25. Kasmaiee, S., Kasmaiee, S. & Homayounpour, M. Correcting spelling mistakes in Persian texts with rules and deep learning methods. *Sci. Rep.* **13**, 19945. https://doi.org/10.1038/s41598-023-47295-2 (2023).
26. Kasmaiee, S. & Tadjfar, M. Non-circular pressure swirl nozzles injecting into stagnant air. *Int. J. Multiph. Flow* **175**, 104798. https://doi.org/10.1016/j.ijmultiphaseflow.2024.104798 (2024).
27. Kasmaiee, S. & Tadjfar, M. Elliptical pressure swirl jet issuing into stagnant air. *Phys. Fluids* https://doi.org/10.1063/5.0198105 (2024).
28. Ruiz, M. C. & D'Ambrosio, D. Aerodynamic optimization and analysis of quadrotor blades operating in the Martian atmosphere. *Aerosp. Sci. Technol.* **132**, 108047. https://doi.org/10.1016/j.ast.2022.108047 (2023).
29. Nissim, L. *et al.* Machine learning based on computational fluid dynamics enables geometric design optimisation of the NeoVAD blades. *Sci. Rep.* **13**, 7183. https://doi.org/10.1038/s41598-023-33708-9 (2023).
30. Kasmaiee, S., Tadjfar, M. & Kasmaiee, S. Optimization of blowing jet performance on wind turbine airfoil under dynamic stall conditions using active machine learning and computational intelligence. *Arab. J. Sci. Eng.* https://doi.org/10.1007/s13369-023-07892-9 (2023).
31. Nejad, P. G. & Zenouz, R. T. Axial compressor blades tip leakage flow control using natural aspiration. *Sci. Rep.* **13**, 22567. https://doi.org/10.1038/s41598-023-49391-9 (2023).
32. Kasmaiee, S., Tadjfar, M. & Kasmaiee, S. Investigation of the impact of blowing jet on the dynamic stall of NACA0012. *J. Appl. Comput. Sci. Mech.* **34**, 1–20. https://doi.org/10.22067/jacsm.2022.74008.1074 (2022).
33. Kasmaiee, S., Tadjfar, M. & Kasmaiee, S. Investigation of suction jet parameters in flow control of dynamic stall. *J. Appl. Comput. Sci. Mech.* **32**, 181–200. https://doi.org/10.22067/jacsm.2021.71526.1042 (2021).
34. Kasmaiee, S., Tadjfar, M. & Kasmaiee, S. Machine learning-based optimization of a pitching airfoil performance in dynamic stall conditions using a suction controller. *Phys. Fluids* https://doi.org/10.1063/5.0164437 (2023).
35. Salah, D. A. H., Nosier, M.A.E.-R. & Hamed, A. M. Investigation of the performance of a horizontal-axis dual rotor wind turbine. *Sci. Rep.* **14**, 6208. https://doi.org/10.1038/s41598-024-55844-6 (2024).

36. Patankar, S. V. *Numerical Heat Transfer and Fluid Flow, Series in Computational Methods in Mechanics and Thermal Sciences* (CRC Press, 1980).
37. Versteeg, H. K. & Malalasekera, W. *An introduction to computational fluid dynamics: the finite method* (Pearson education, 2007).
38. Menter, F. Zonal two equation kw turbulence models for aerodynamic flows. in *23rd fluid dynamics, plasmadynamics, and lasers conference 2906* (1993). https://doi.org/10.2514/6.1993-2906.
39. Manufacturer's website KV29 industrial blade. https://store.tmotor.com/goods.php?id=891
40. Wang, P. *et al.* Investigation of the aerodynamic characteristics of horizontal axis wind turbine using an active flow control method via boundary layer suction. *Renew Energy* **198**, 1032–1048. https://doi.org/10.1016/j.renene.2022.08.075 (2022).
41. Kasmaiee, S. & Tadjfar, M. Influence of injection angle on liquid jet in crossflow. *Int. J. Multiph. Flow* https://doi.org/10.1016/j.ijmultiphaseflow.2022.104128 (2022).
42. Kasmaiee, S. & Tadjfar, M. Experimental study of the injection angle impact on the column waves: wavelength, frequency and drop size. *Exp. Therm. Fluid Sci.* https://doi.org/10.1016/j.expthermflusci.2023.110989 (2023).
43. Kasmaiee, S., Tadjfar, M., Kasmaiee, S. & Ahmadi, G. Linear stability analysis of surface waves of liquid jet injected in transverse gas flow with different angles. *Theor. Comput. Fluid Dyn.* https://doi.org/10.1007/s00162-024-00685-2 (2024).

Author contributions

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Sa. Kasmaiee, Si.Kasmaiee and A. Farshad. The first draft of the manuscript was written by Sa. Kasmaiee and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding

The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to S.K.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2024